

The background is an abstract composition of various geometric shapes and lines. It features a mix of colors including shades of pink, grey, blue, and yellow. Thin white lines crisscross the entire image, creating a sense of depth and complexity. Some areas have thicker, more solid-colored blocks, while others are more transparent, allowing the underlying colors to show through.

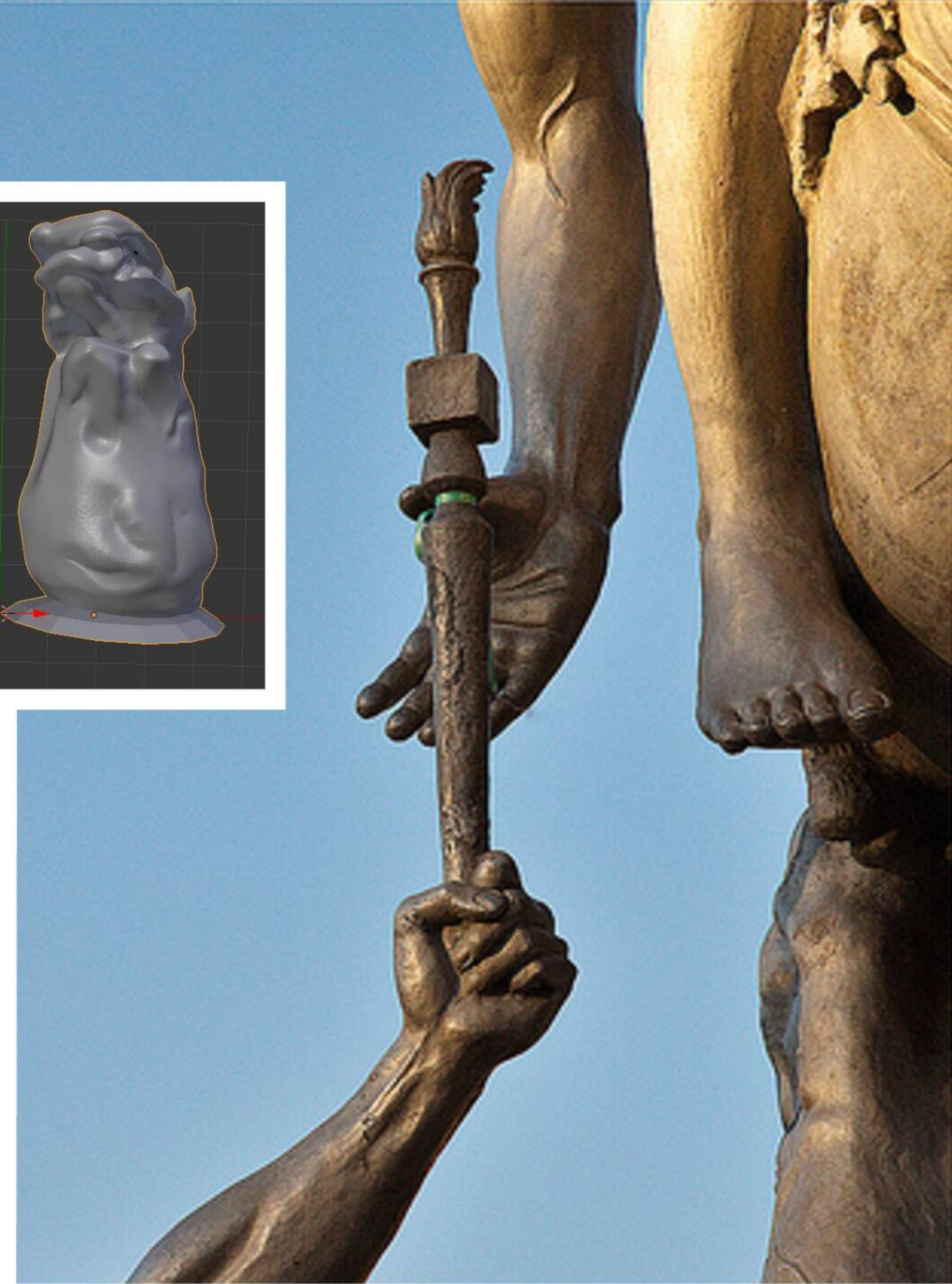
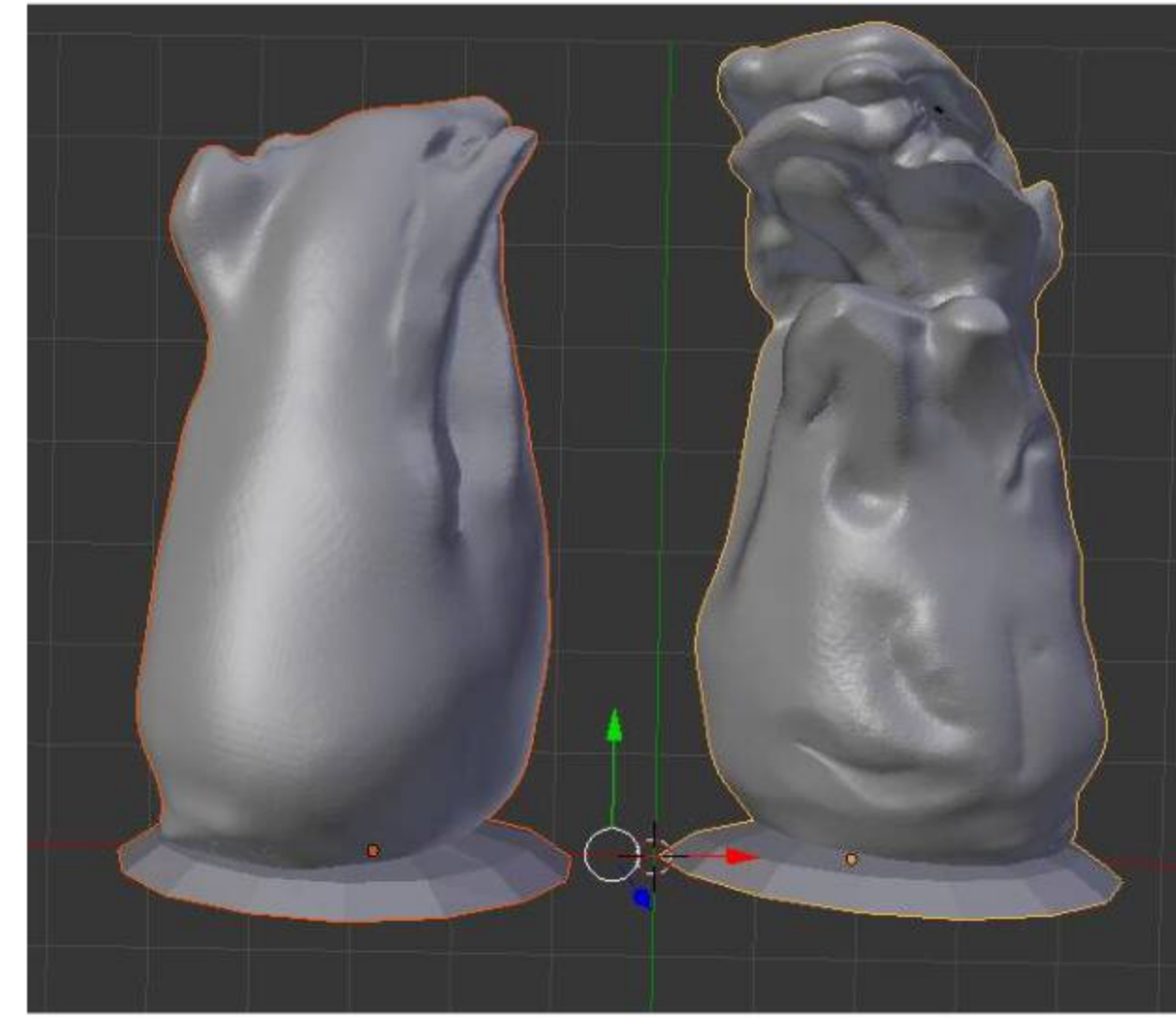
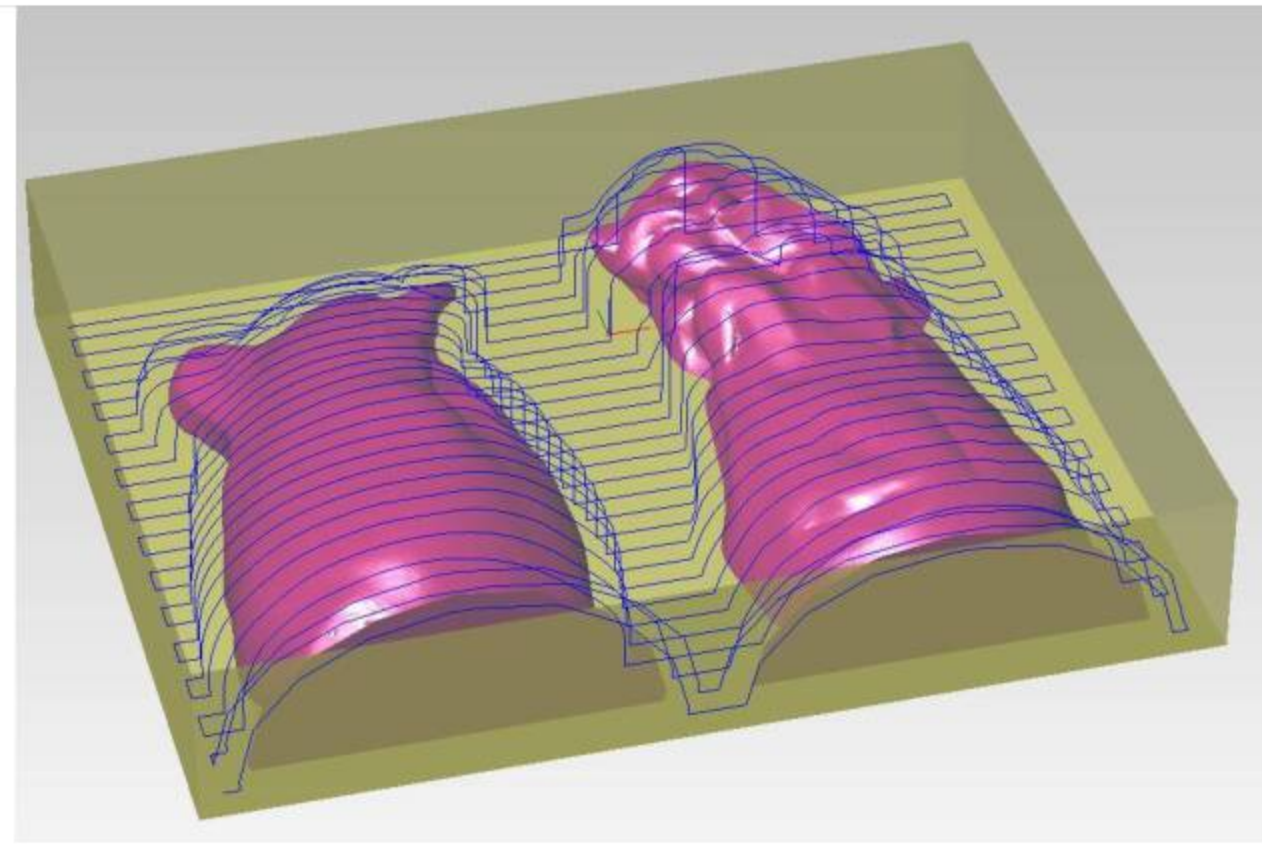
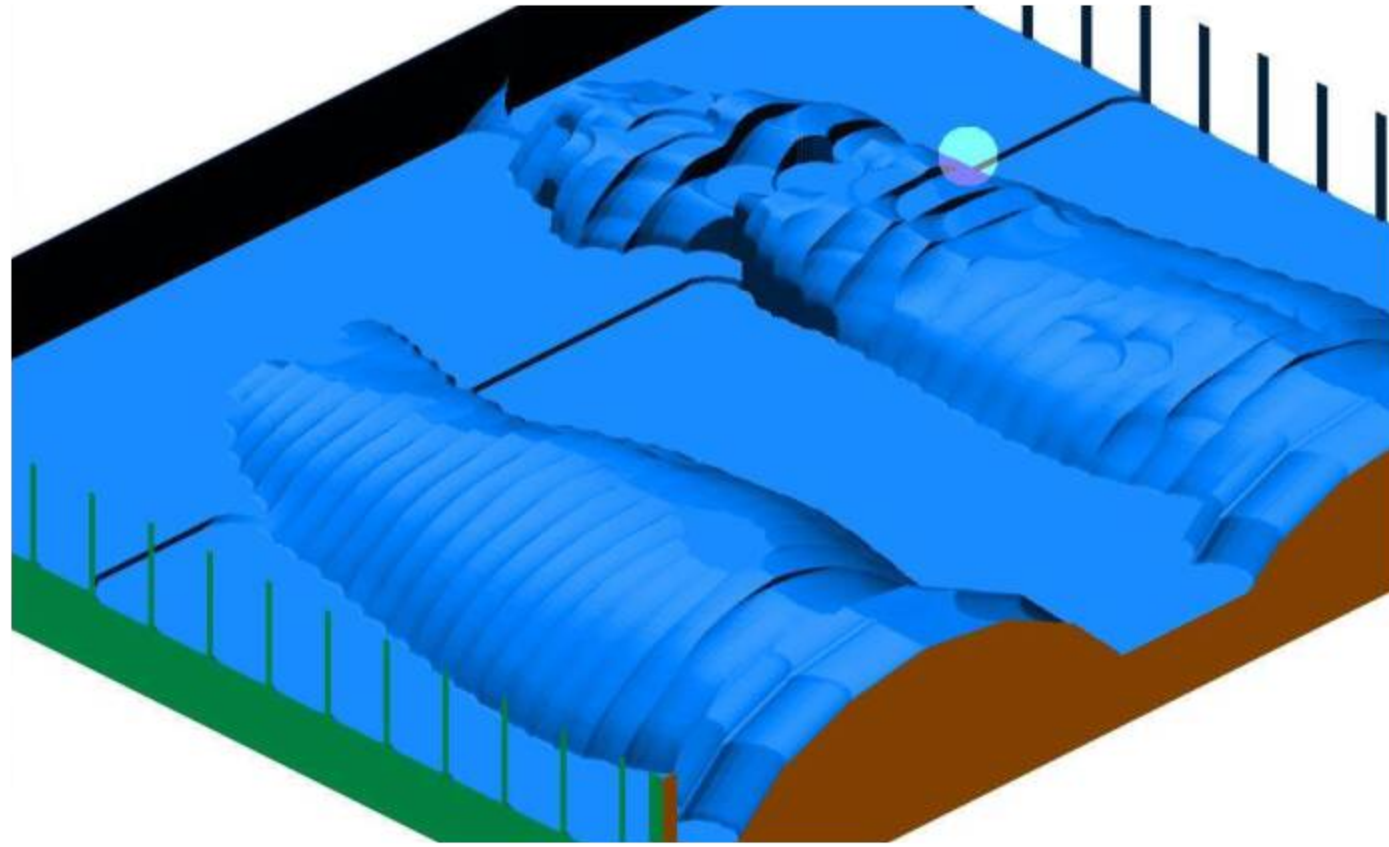
*Javier Gorostiza*

*Dossier Habilidades Digitales*

*[javier.de.gorostiza@gmail.com](mailto:javier.de.gorostiza@gmail.com)*



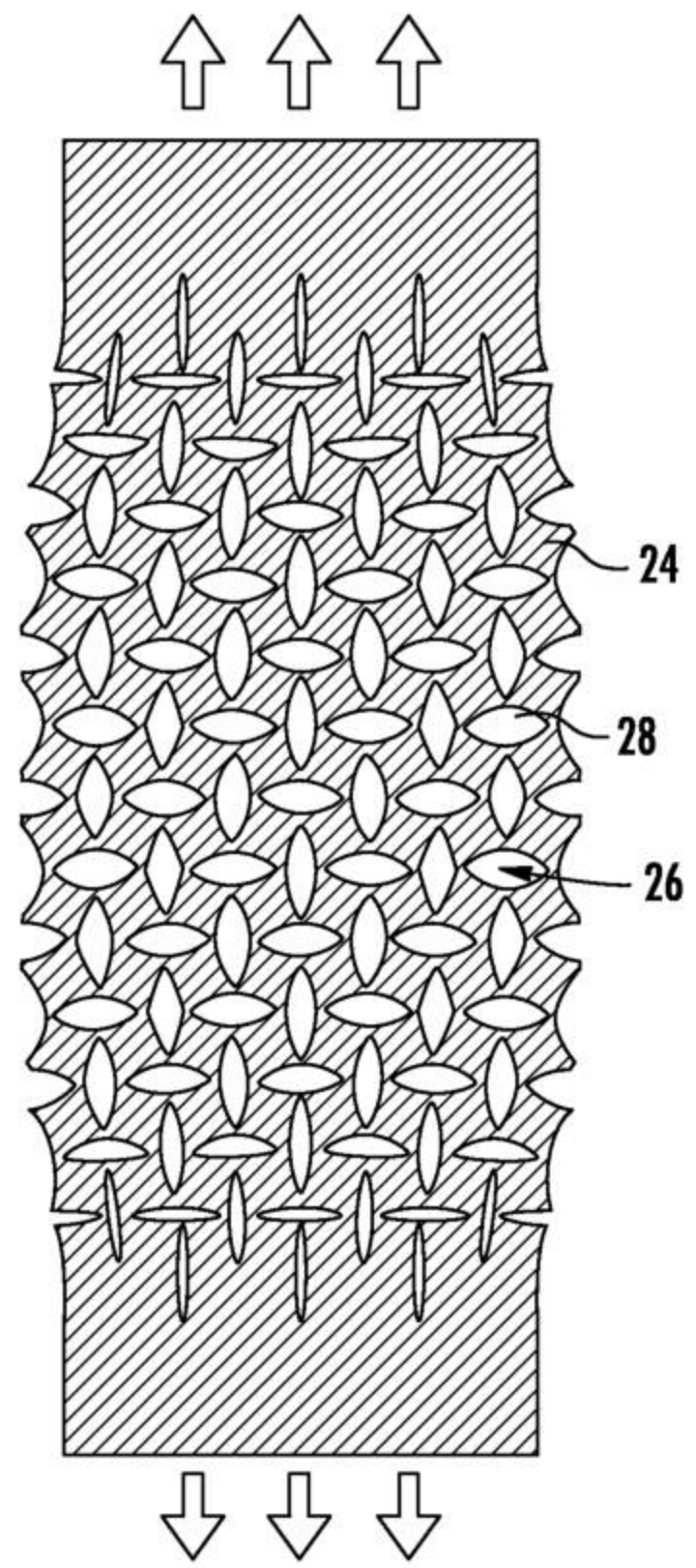
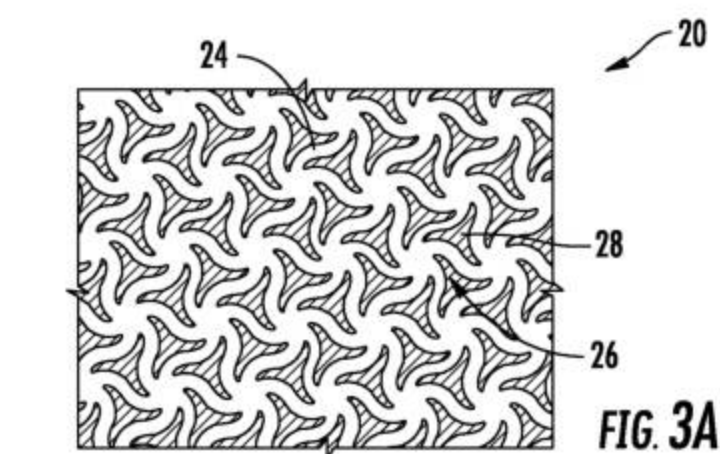
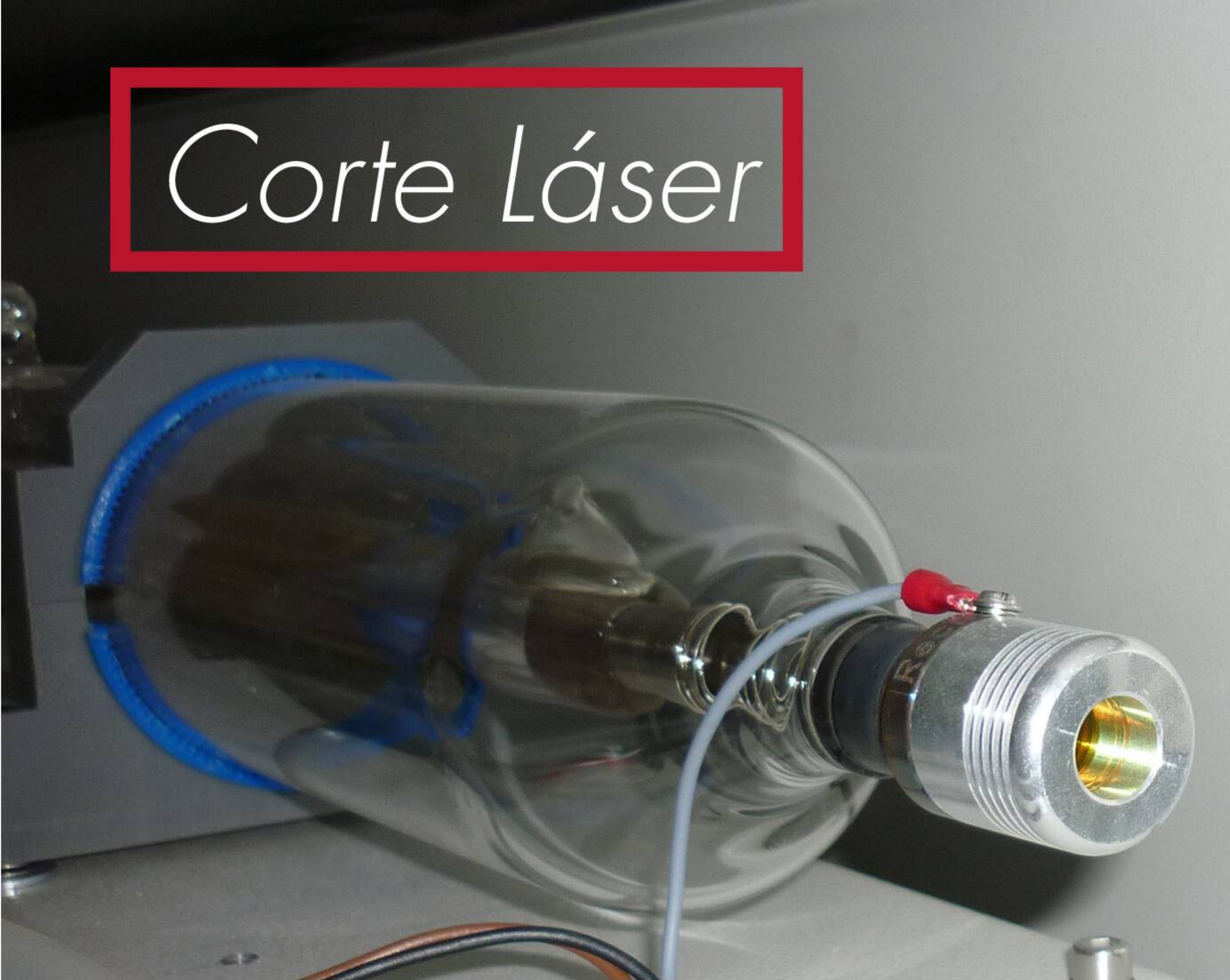
# Mecanizado CNC



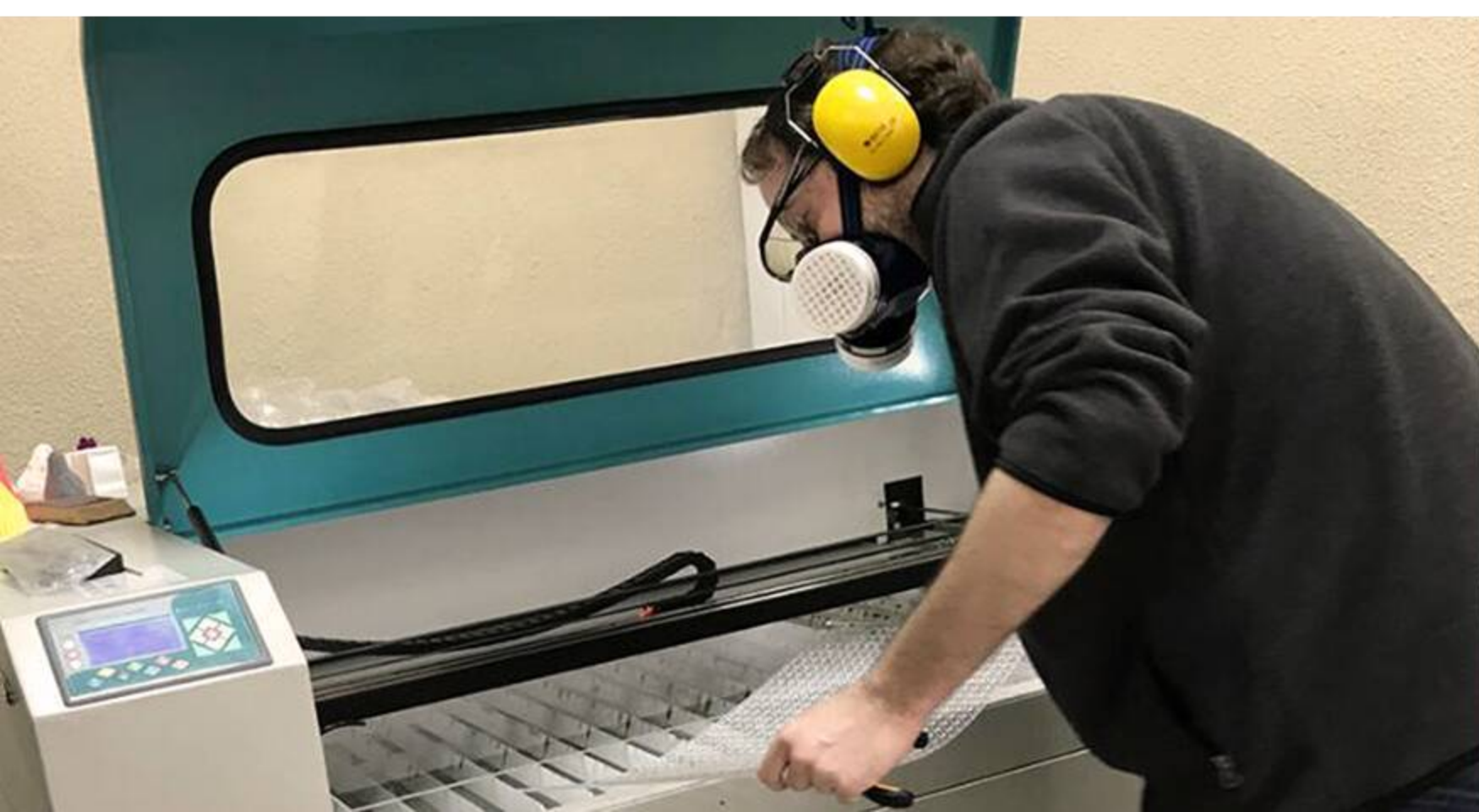
Proyecto de Restauración PIMCD  
'Los portadores de la antorcha'



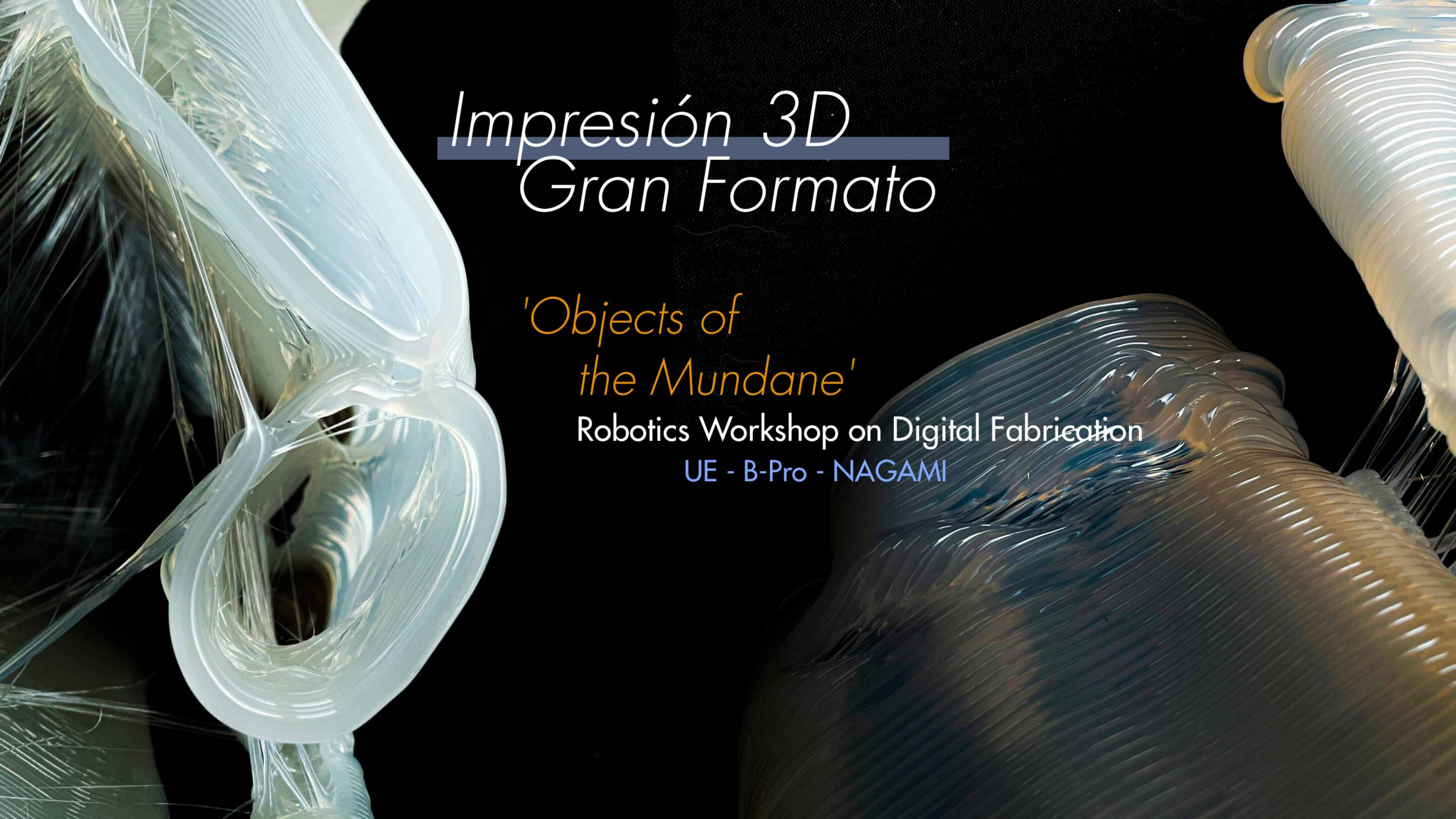
# Corte Láser



Fabricando EPIs durante la emergencia provocada por el SARS-COVID-19







# *Impresión 3D* *Gran Formato*

*'Objects of  
the Mundane'*

Robotics Workshop on Digital Fabrication

UE - B-Pro - NAGAMI









# *Impresión 3D* *Investigación de Materiales*





*'Innova-Docencia  
Proyecto N° 339'*

Positivado en hormigón de piezas  
generativas a partir de moldes  
impresos en FDM.



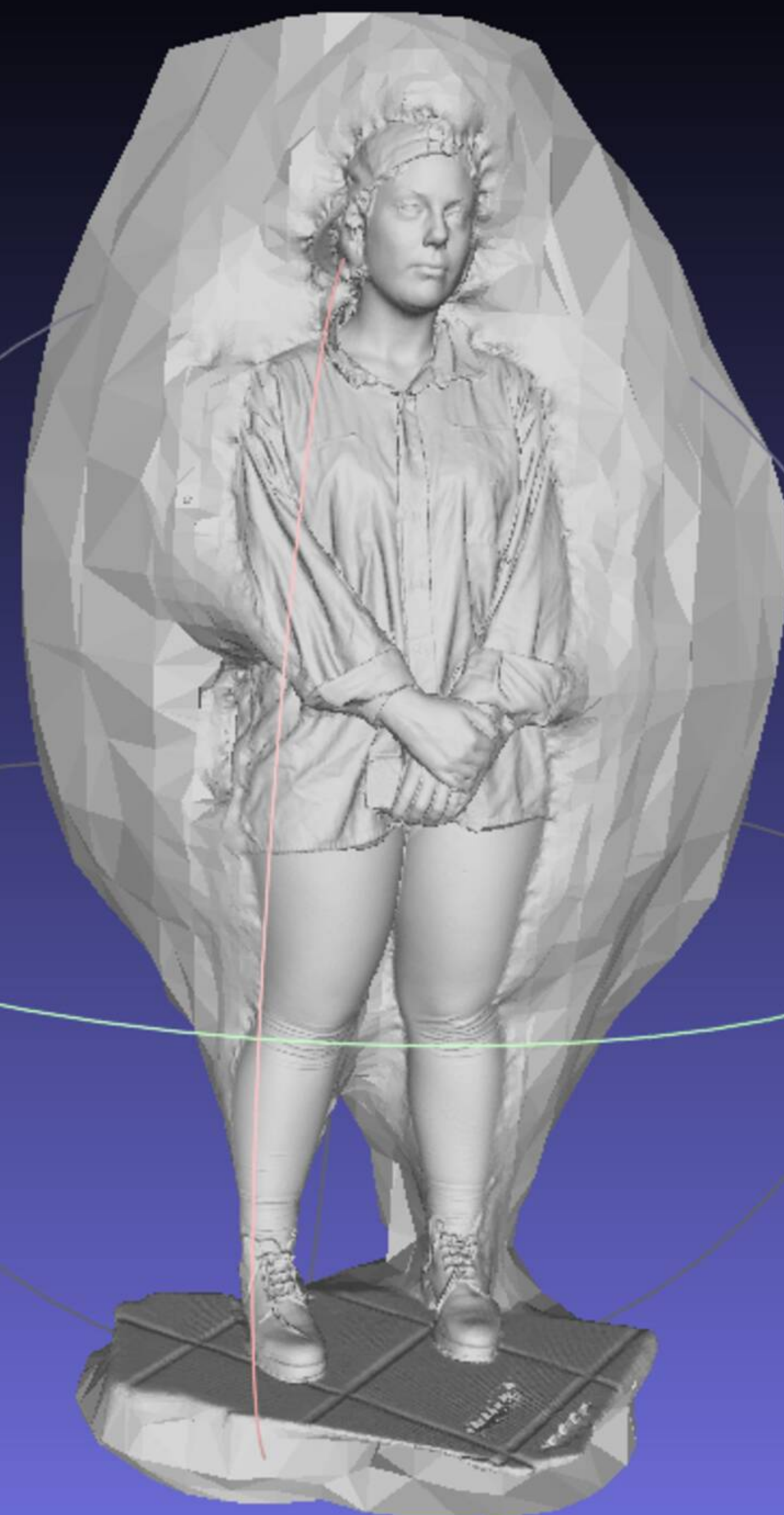
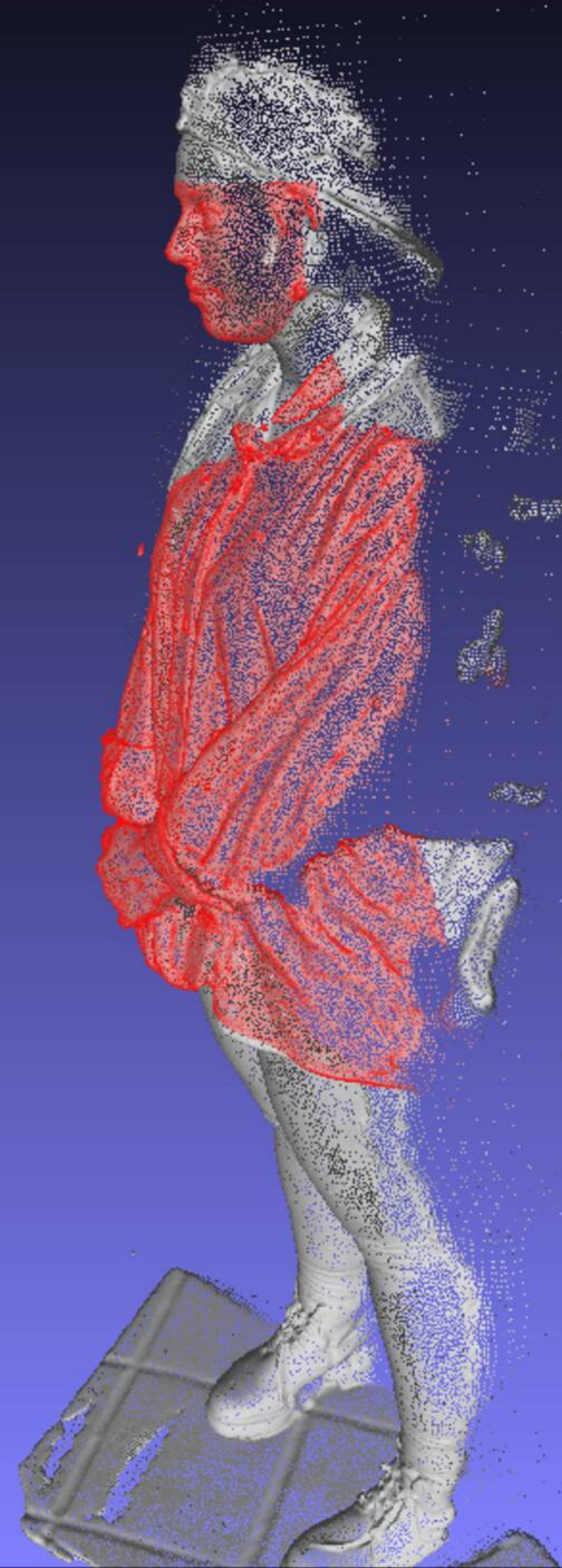


## 'Impresión 3D en cerámica'





# Fotogrametría y *Luz Estructurada*



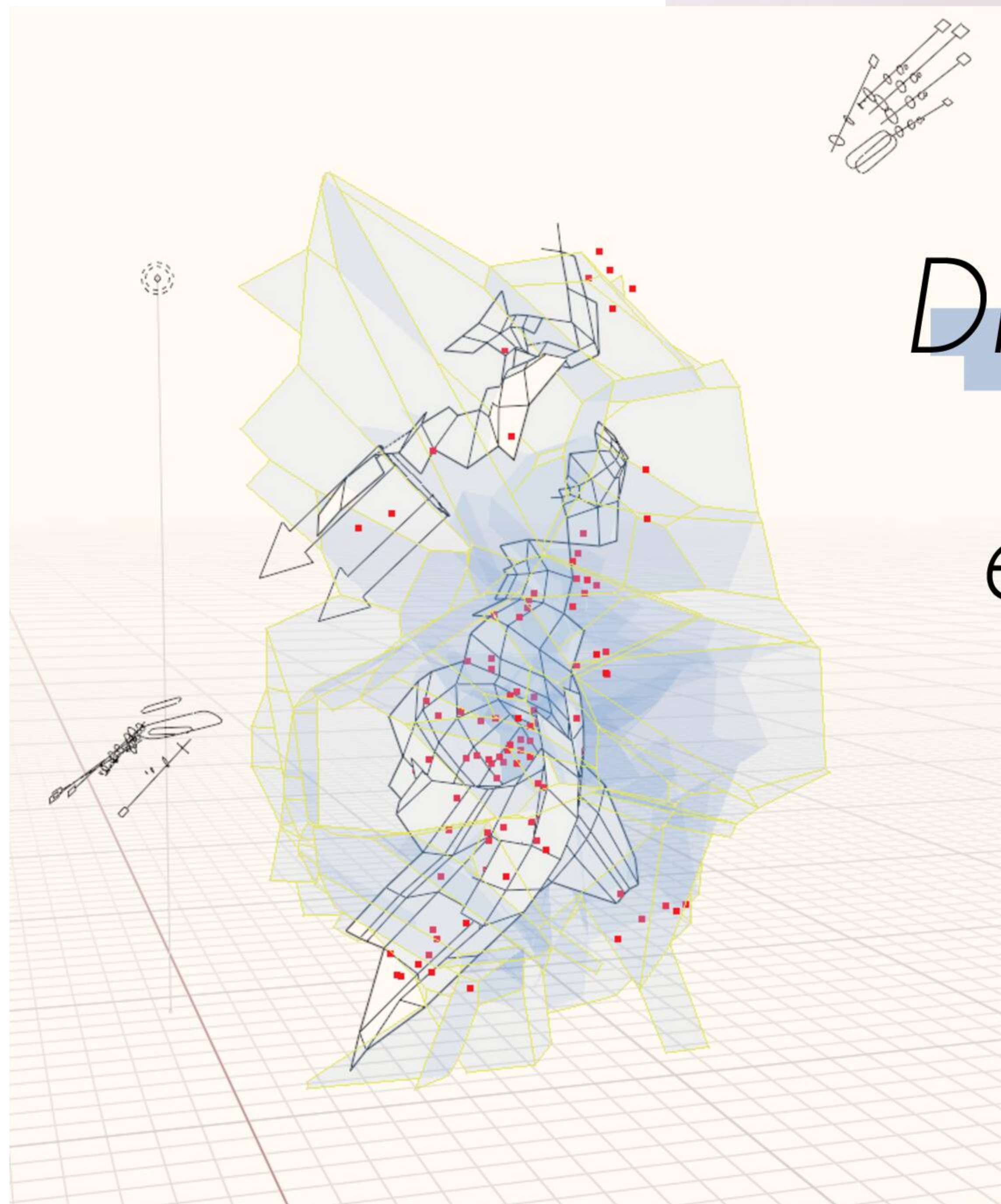




# *Digitalización de Espacios y Edificios*

- *Reality Capture*
- *Meshroom*
- *Agisoft Photoscan*
- *Meshlab*



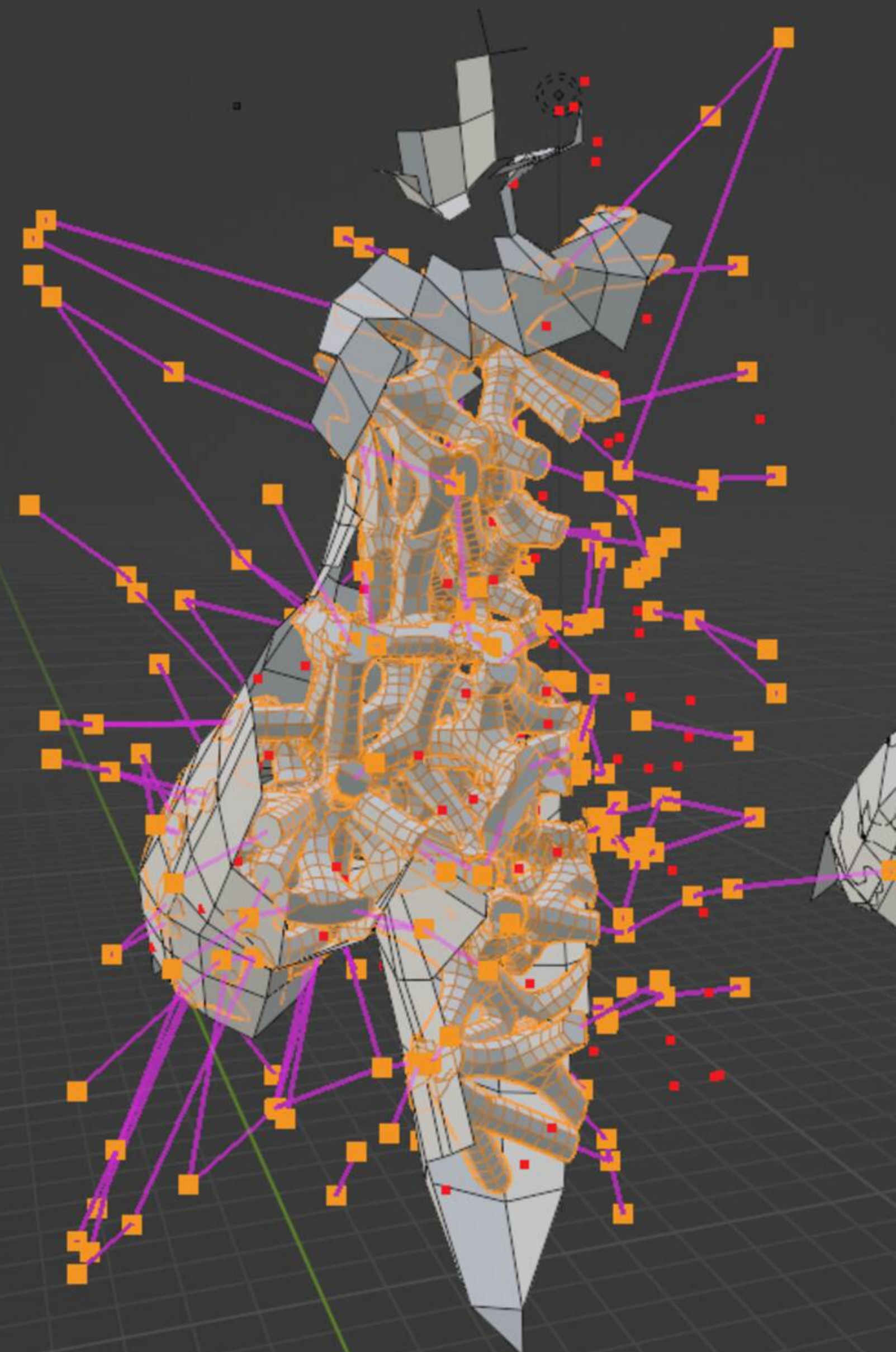
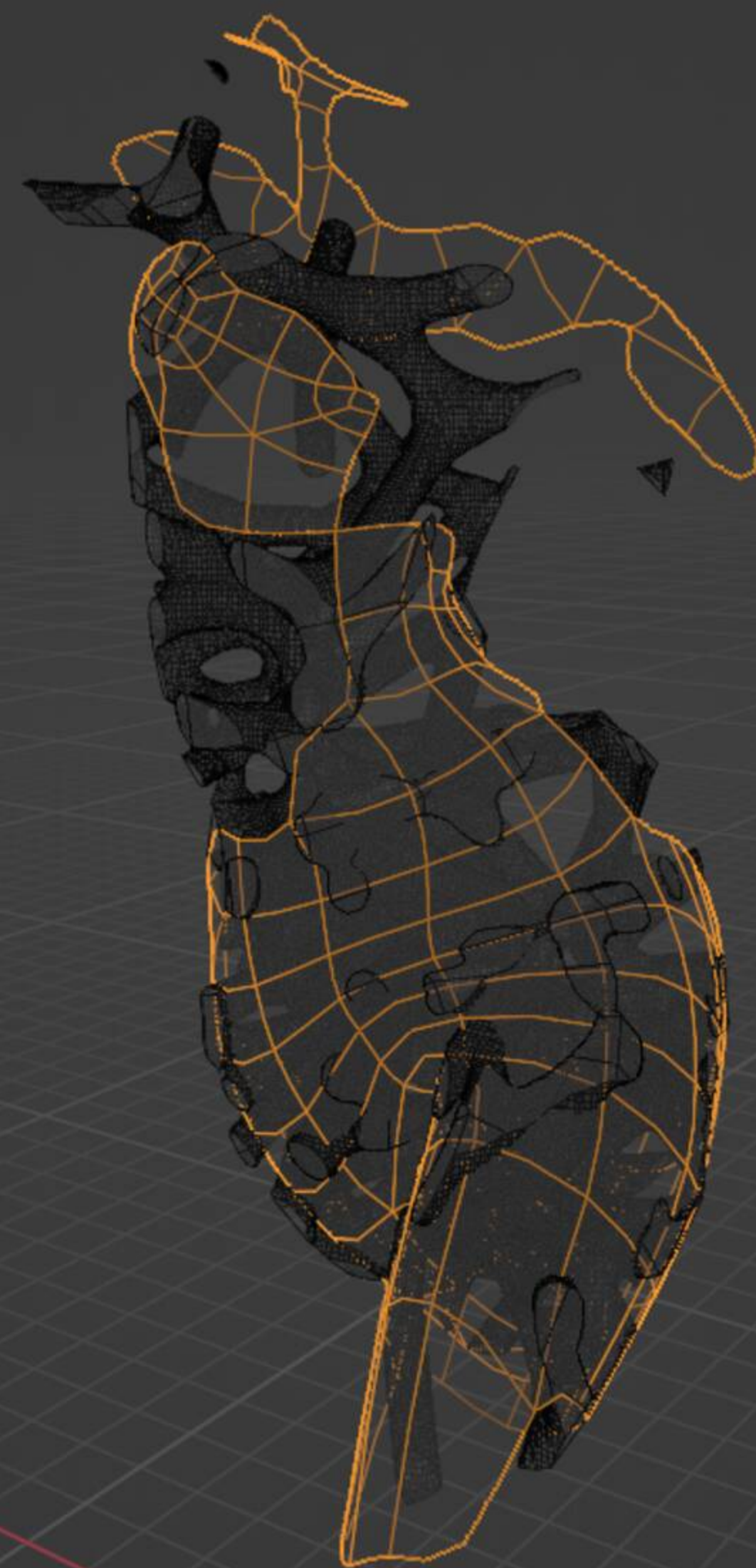
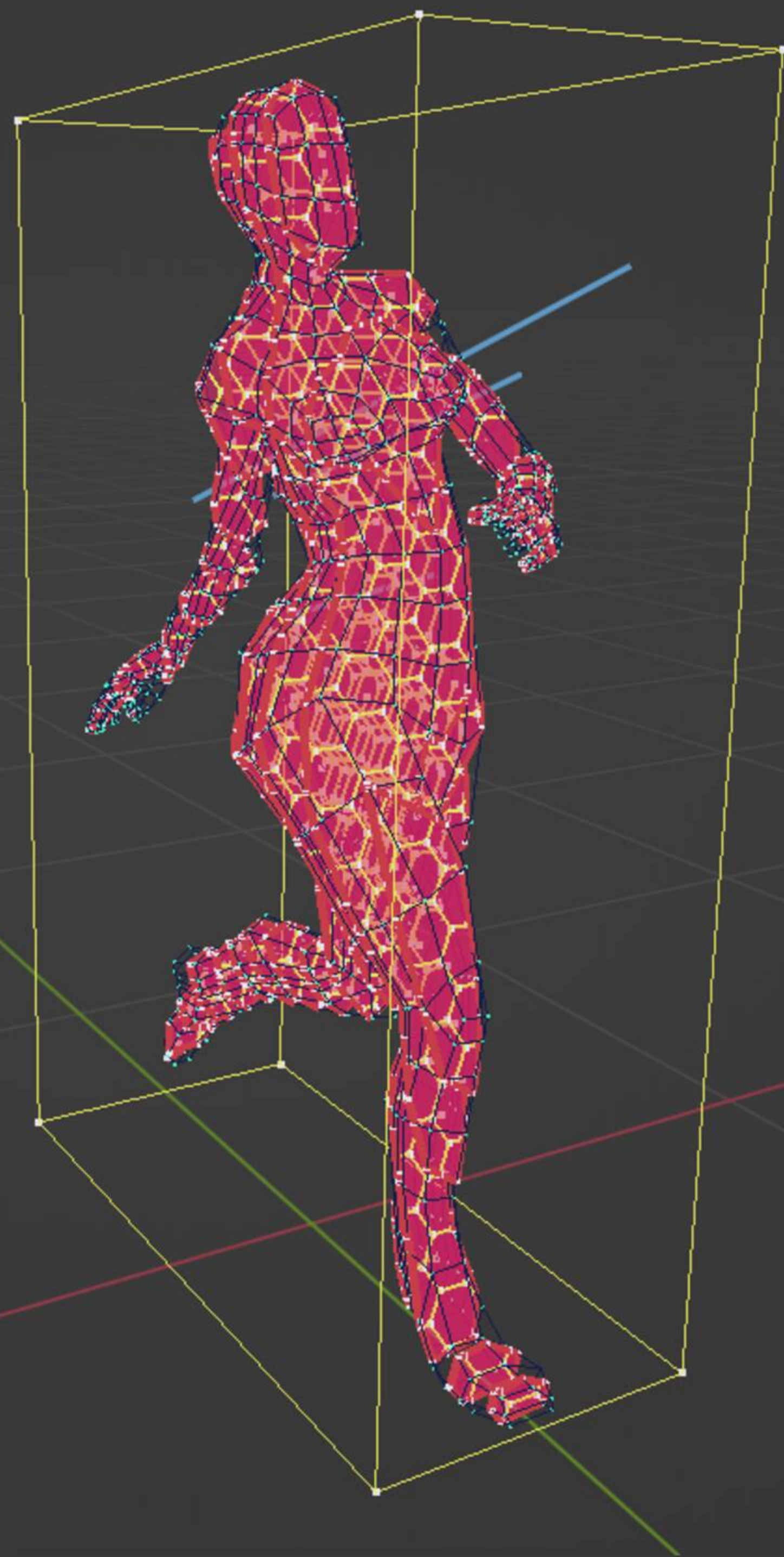
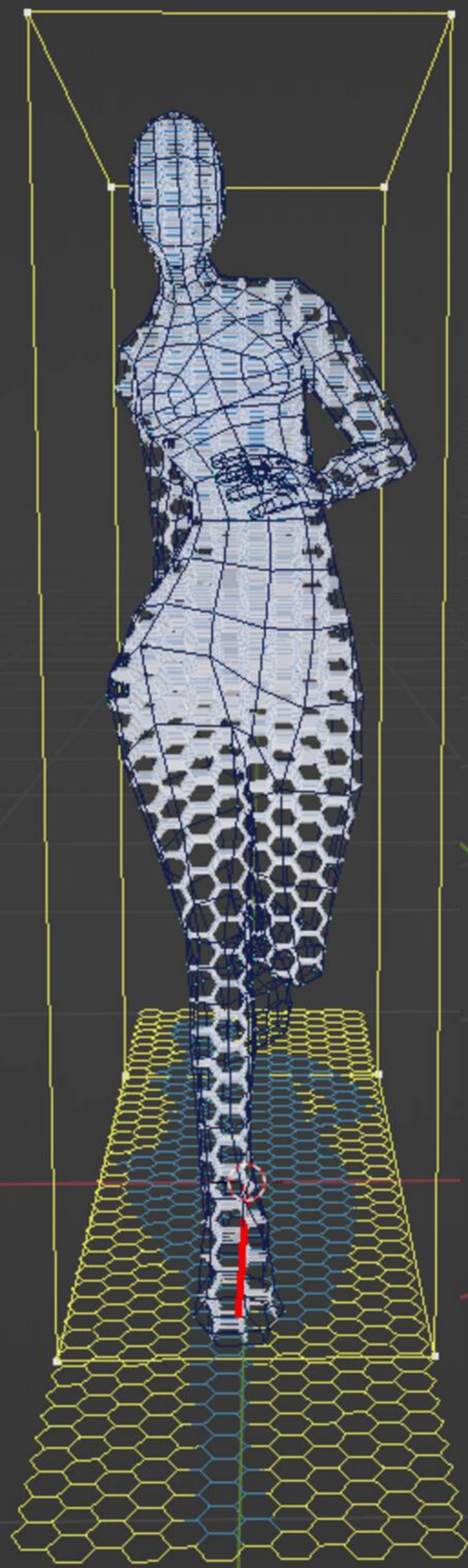


# *Diseño Generativo en Blender*

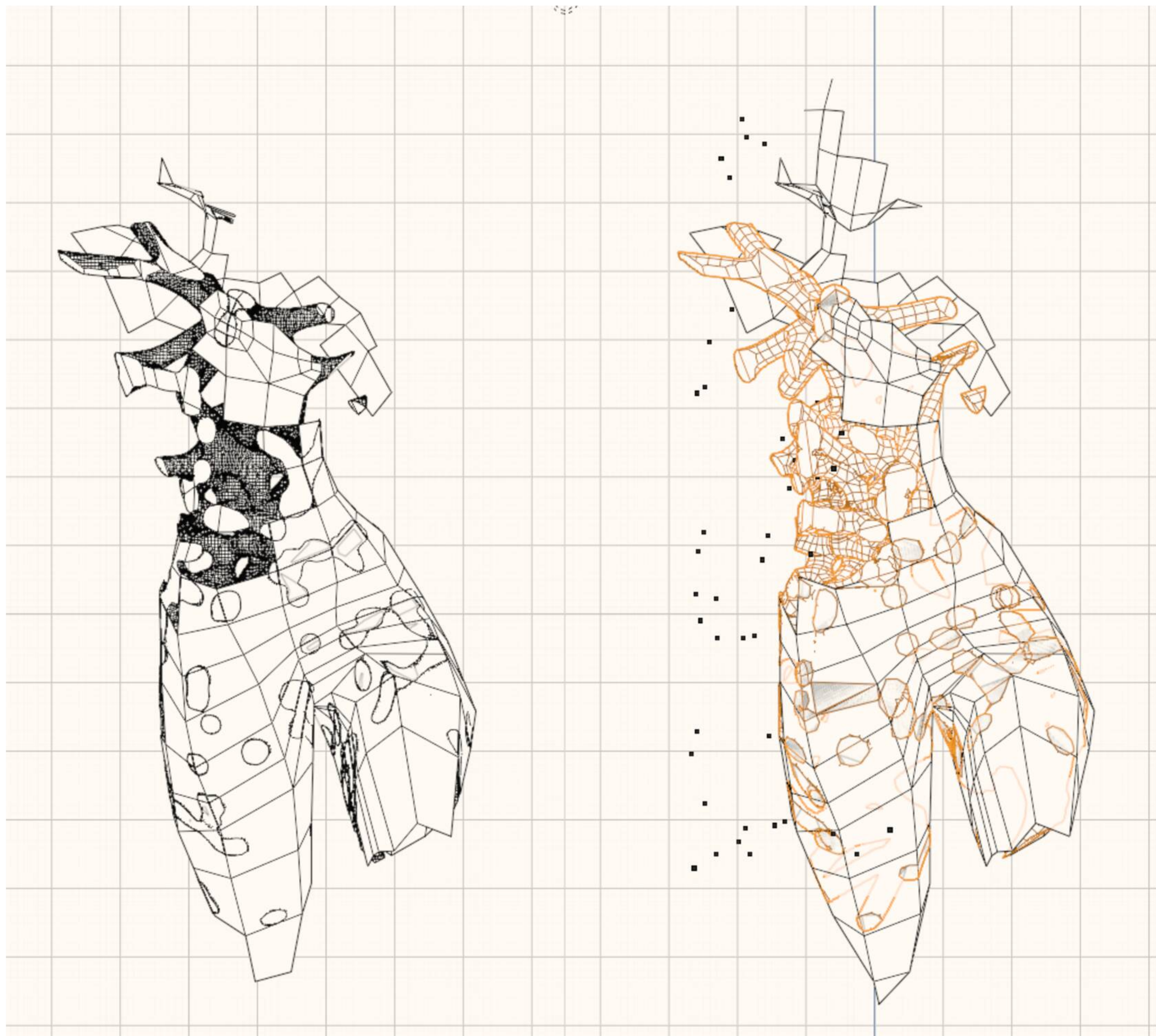
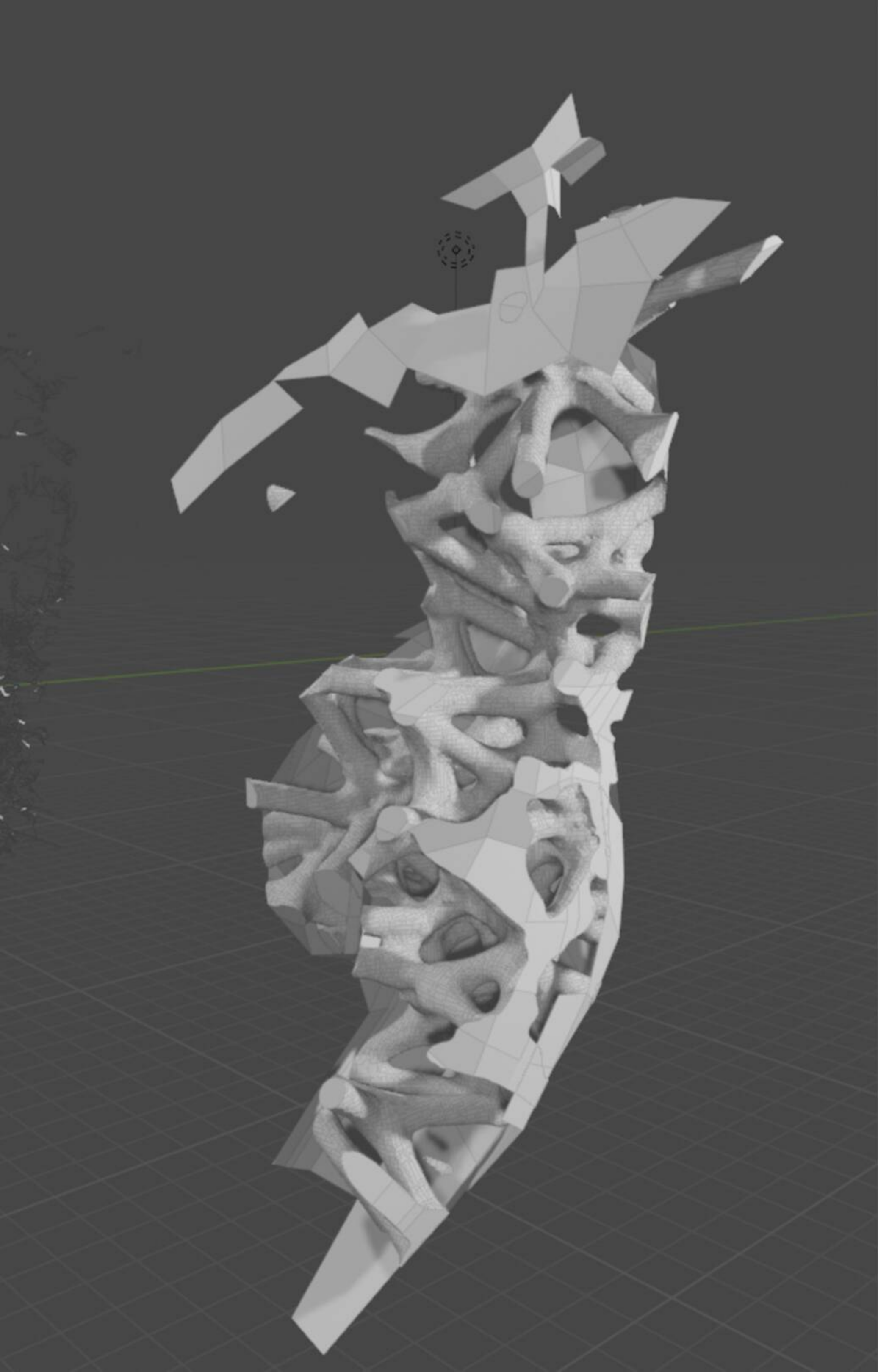
*Impresión en Metal  
certámen CENIM*





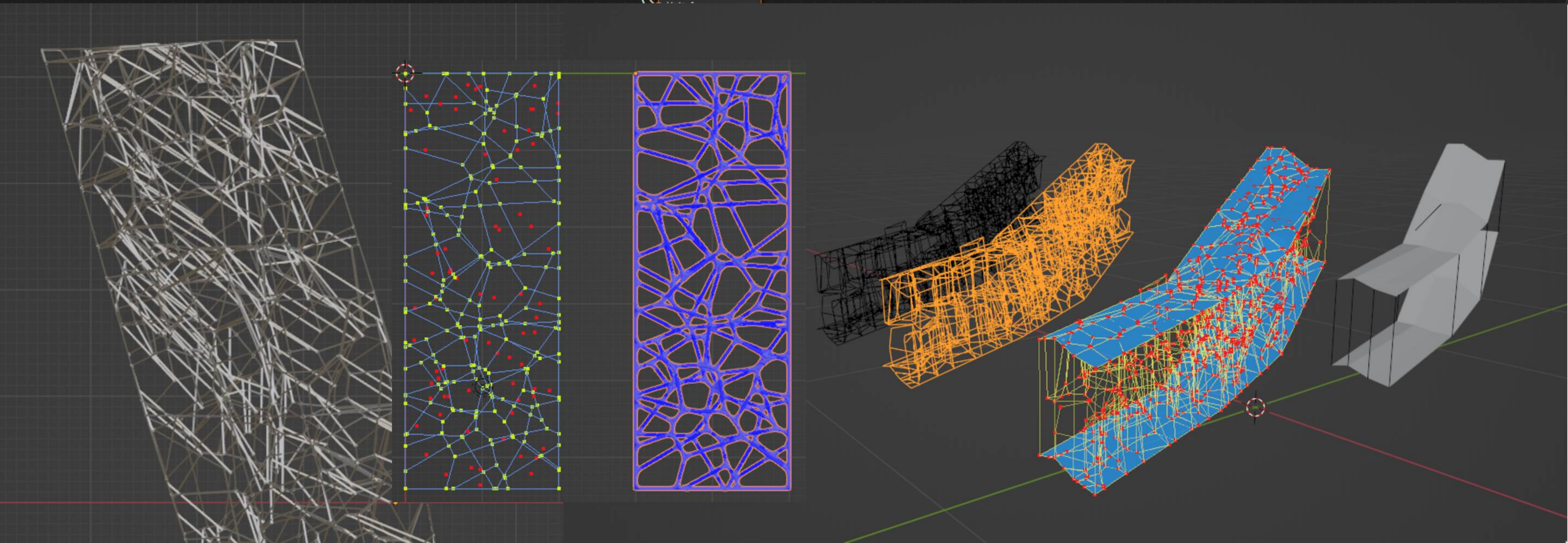
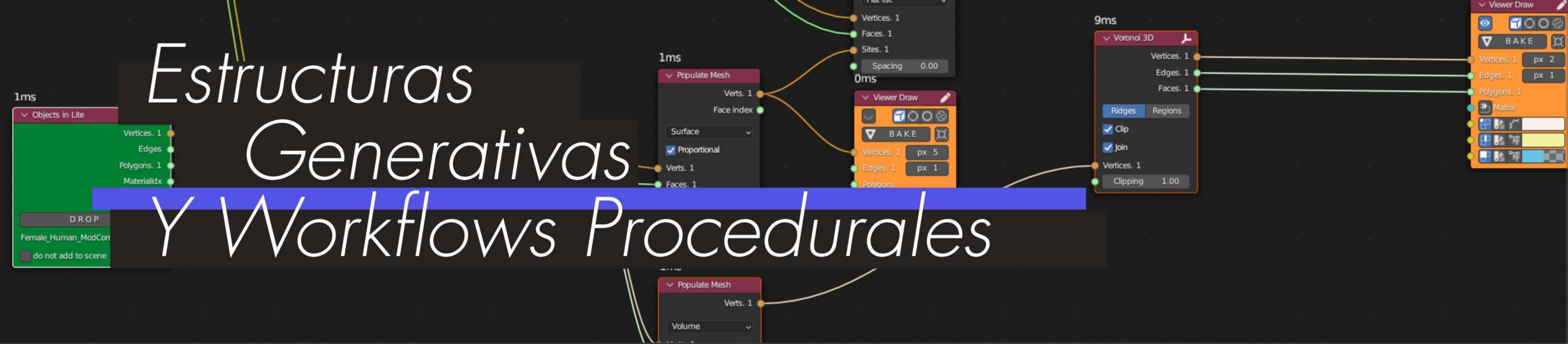




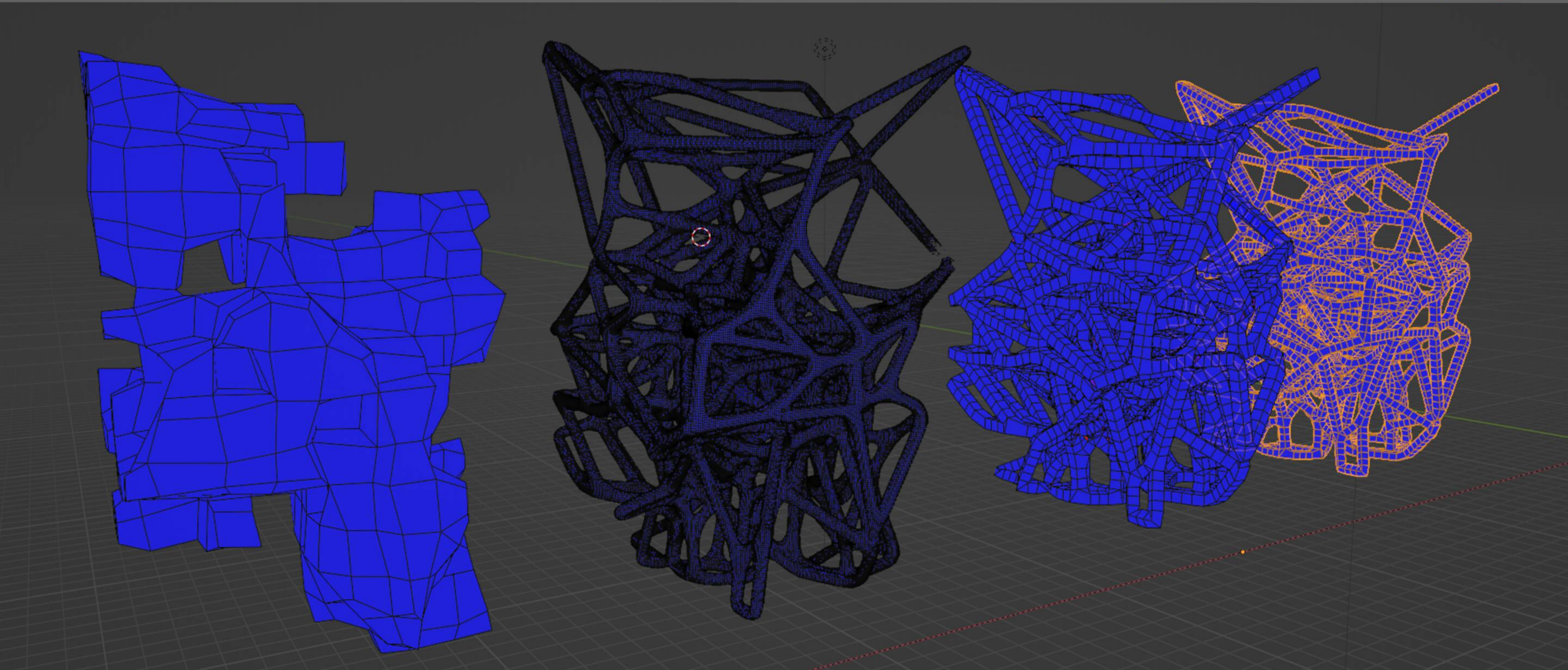
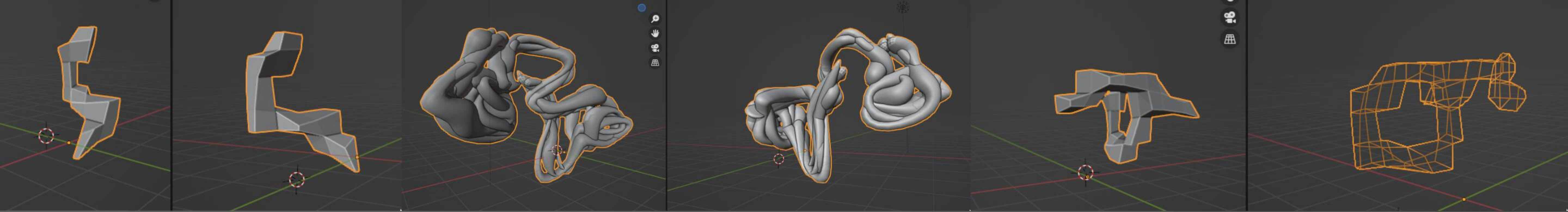




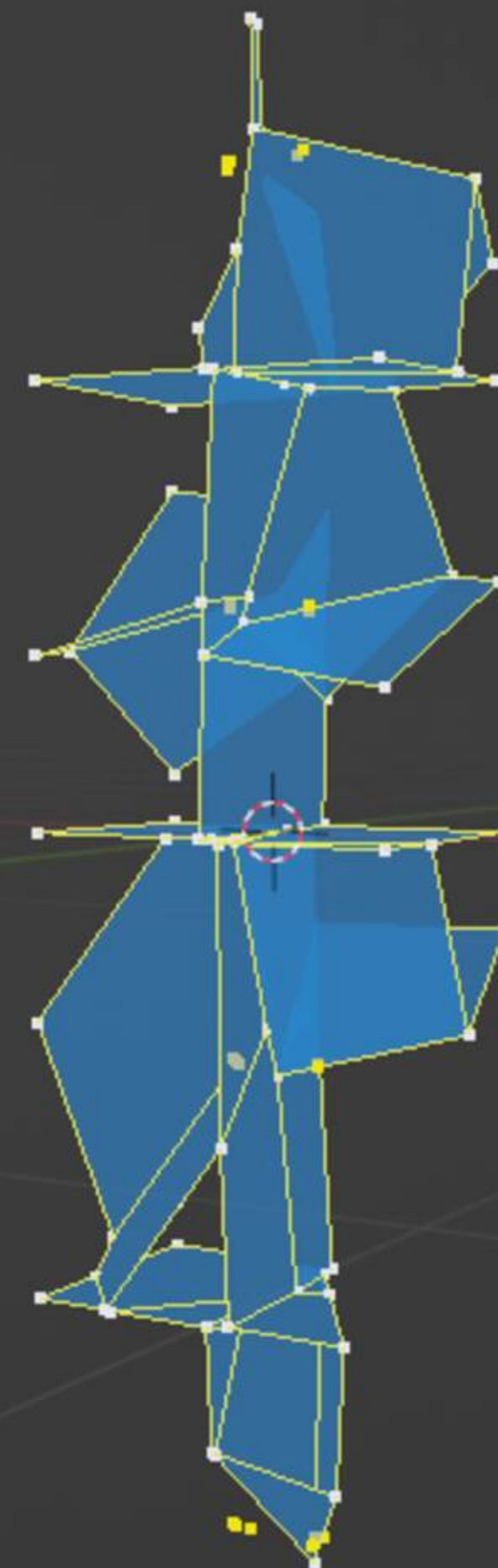
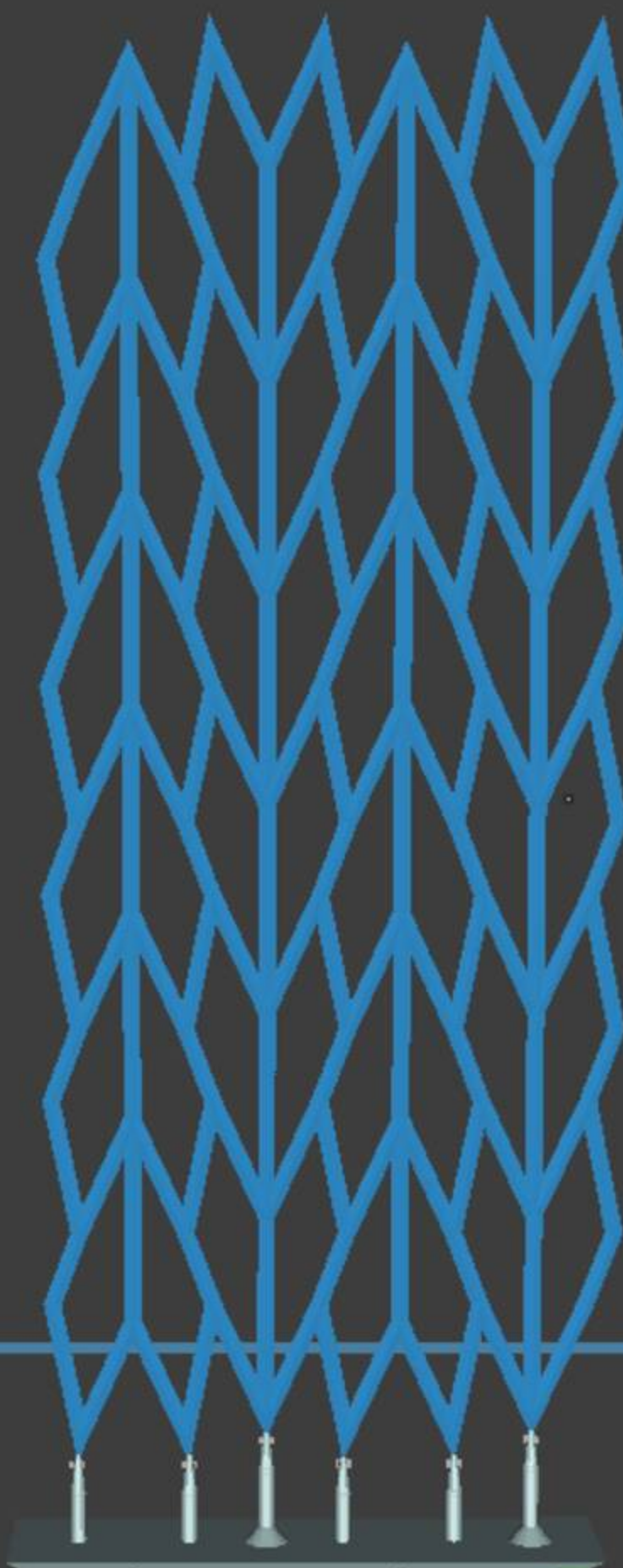
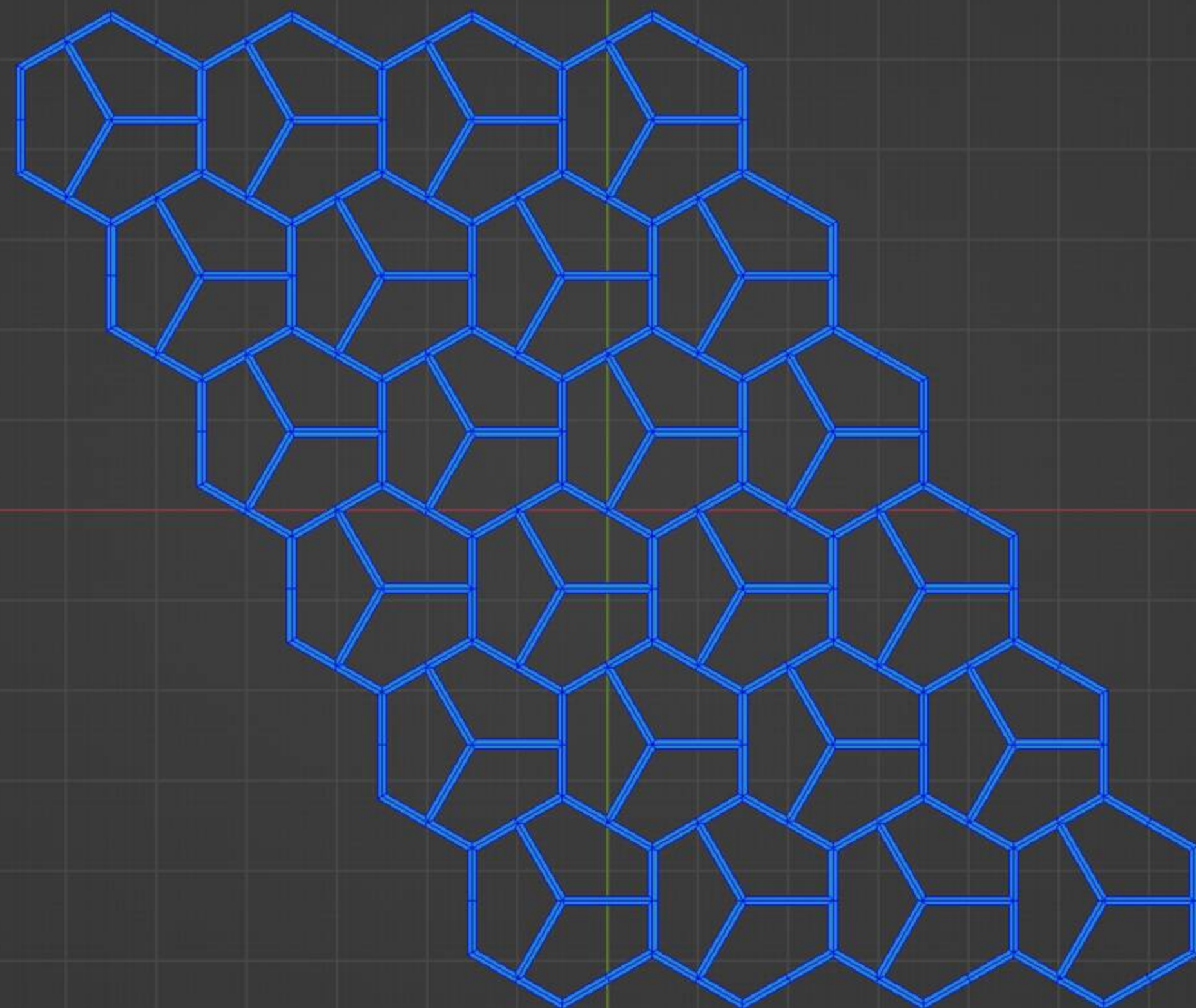
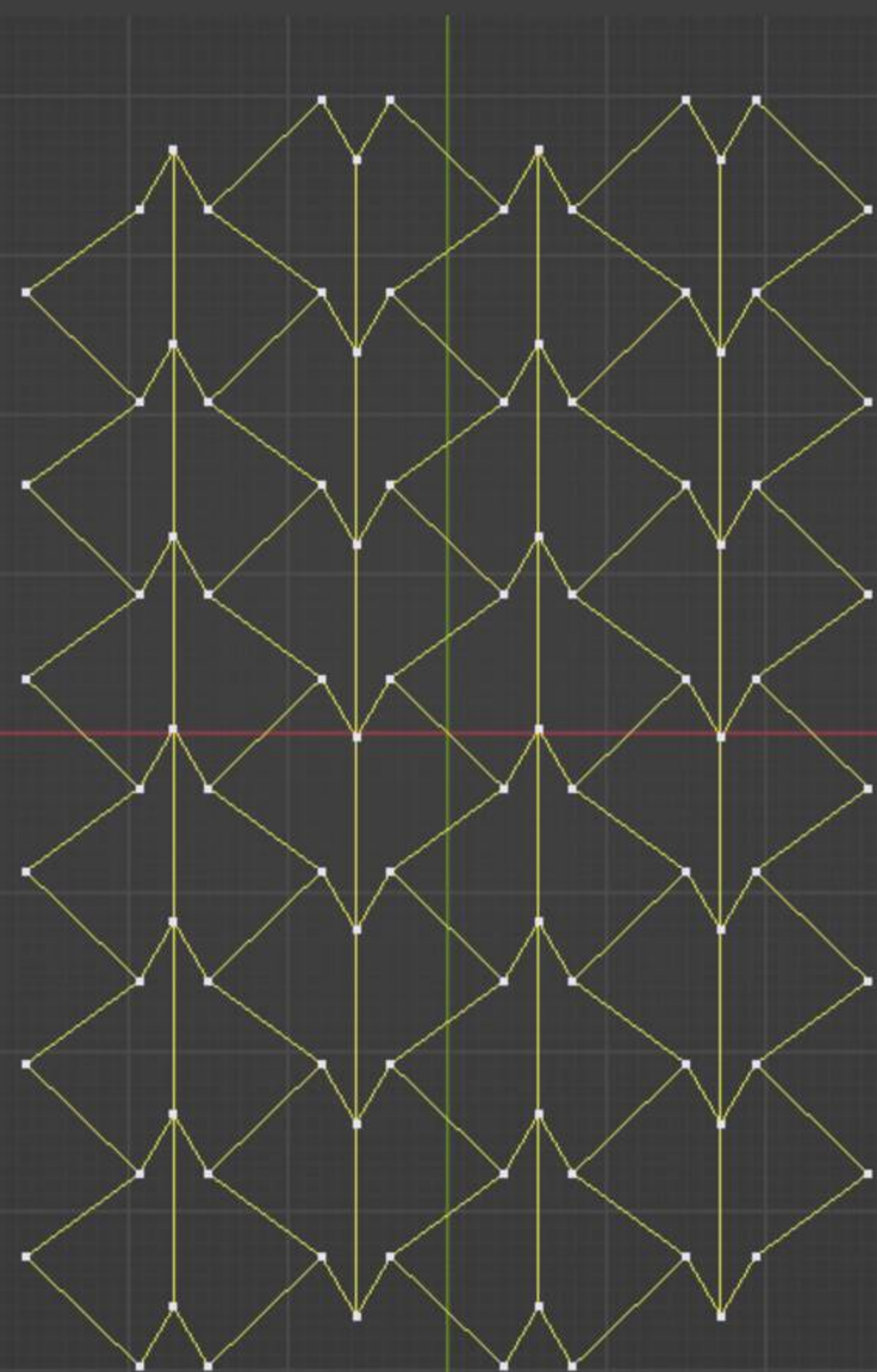
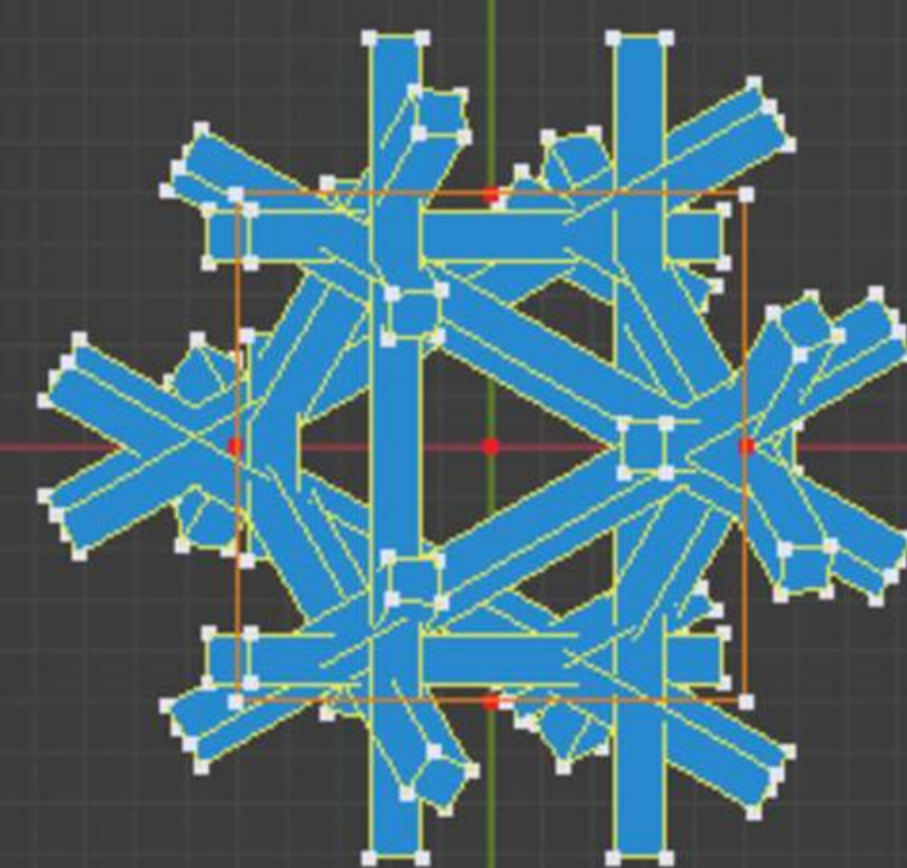
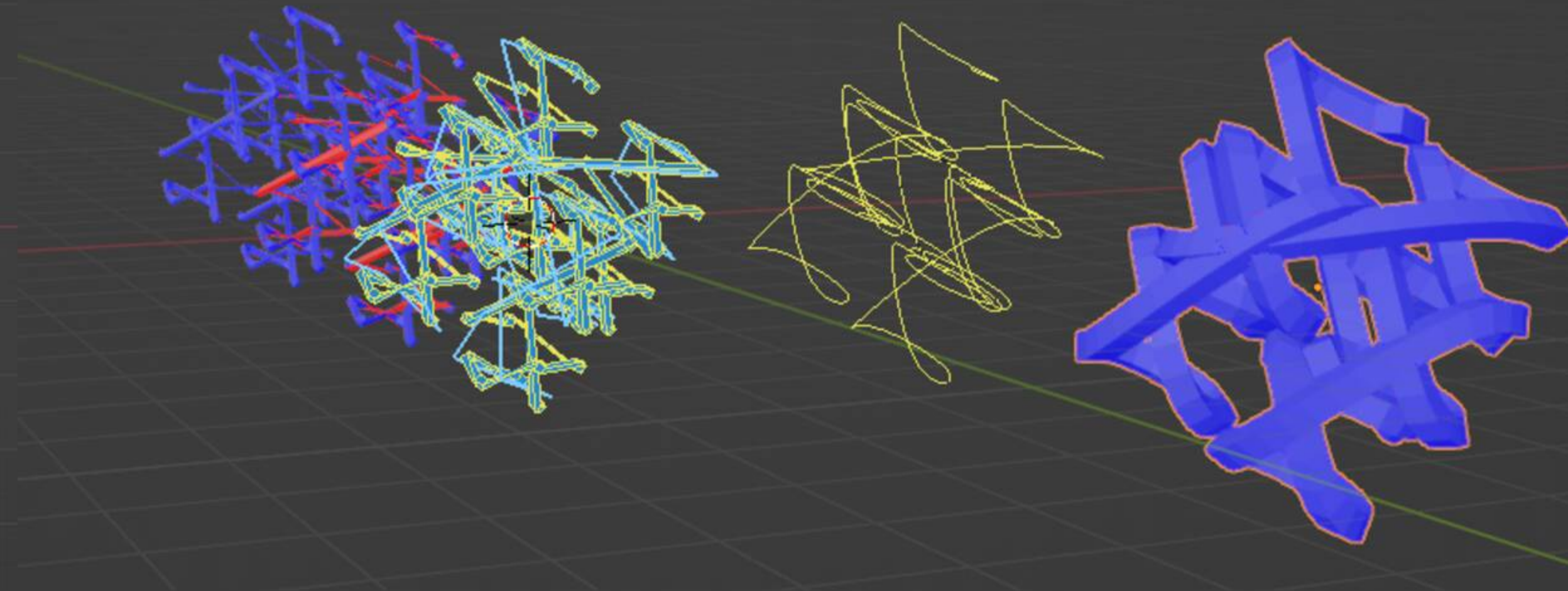
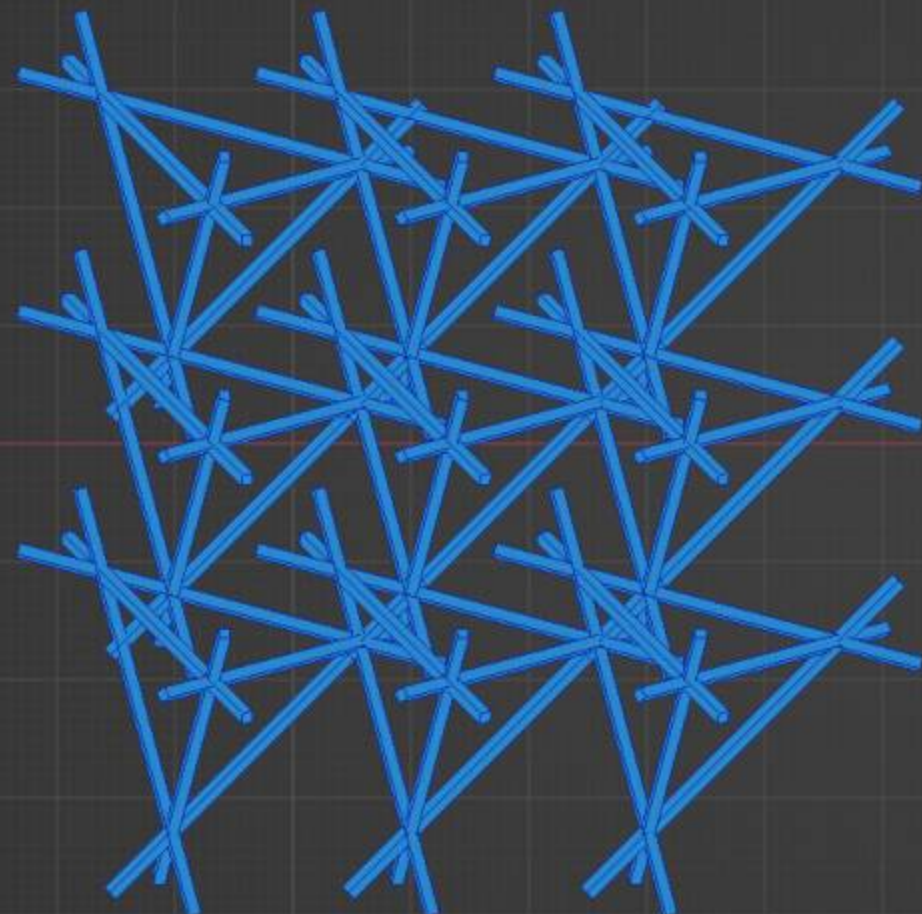
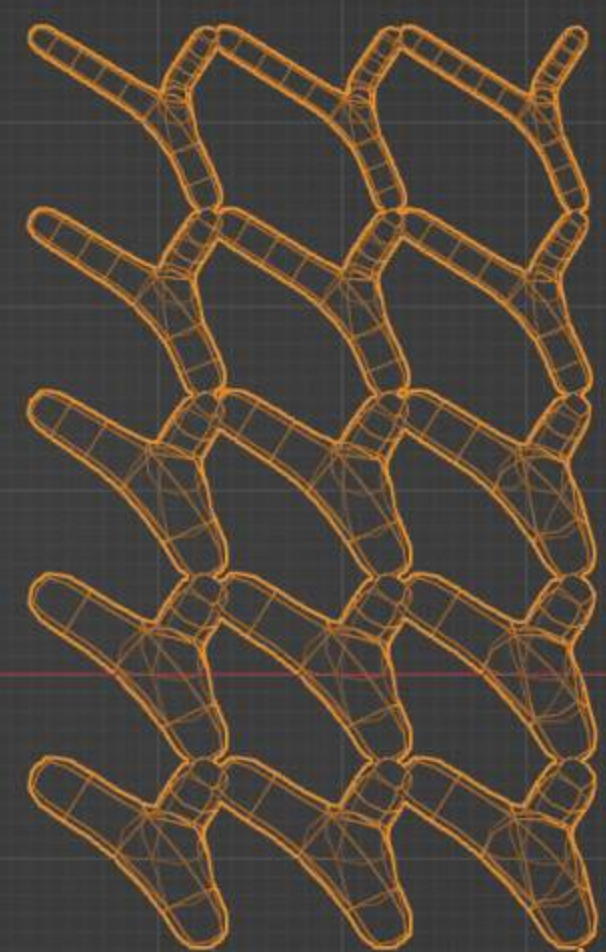
# Estructuras Generativas Y Workflows Procedurales













# CadQuery

## Python Parametric CAD

```
236 ..... name="bottom_holder",
237 ..... color=cq.Color("brown"),
238 ..... loc=cq.Location(cq.Vector(0,0,-height/2))
239 .....
259 ..... .add(pletina, name="pletina", color=cq.Color("lightblue"))
260 .....
261 ..... .constrain("pletina", "Fixed")
262 ..... .constrain("anode_holder/bottom_holder@faces@<Z", "p1
```

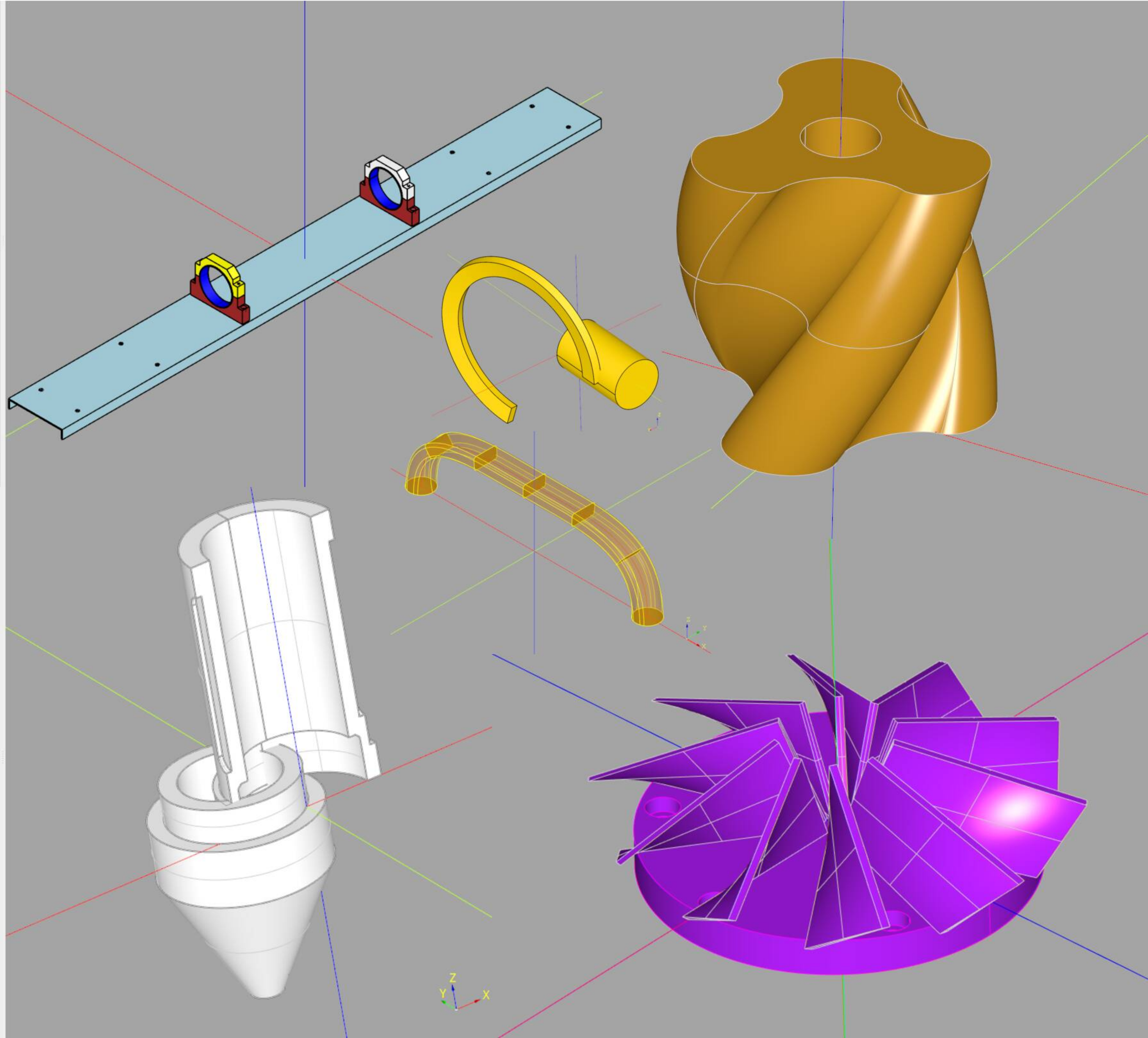
```
cadquery as cq
d_cylinder.head_cylinder import head_cylinder, half_cyl
d_nozzle import head_nozzle
s_nut.lens_nut import lens_nut
s_washer.lens_washer import lens_washer
```

```
ad=(cq.Assembly())
.....add(half_cyl, name="head_cylinder")
.....add(lens_nut, name="lens_nut")
.....add(head_nozzle, name="head_nozzle")
.....add(lens_nut, name="lens_nut")
.....add(lens_washer, name="lens_washer")

.....constrain("head_cylinder", "Fixed")
.....constrain("lens_nut@faces@<Z", "head_cylinder?thread_start", "Po
.....constrain("head_nozzle@faces@>Z[1]", "head_cylinder@faces@<Z", "P
.....constrain("head_nozzle@faces@>Z", "head_cylinder@faces@<Z", "Axi
.....constrain("head_cylinder@faces@<Z", "head_nozzle@faces@>Z", "Poi

.....solve()
.....)
```

```
ect(laser_head)
```



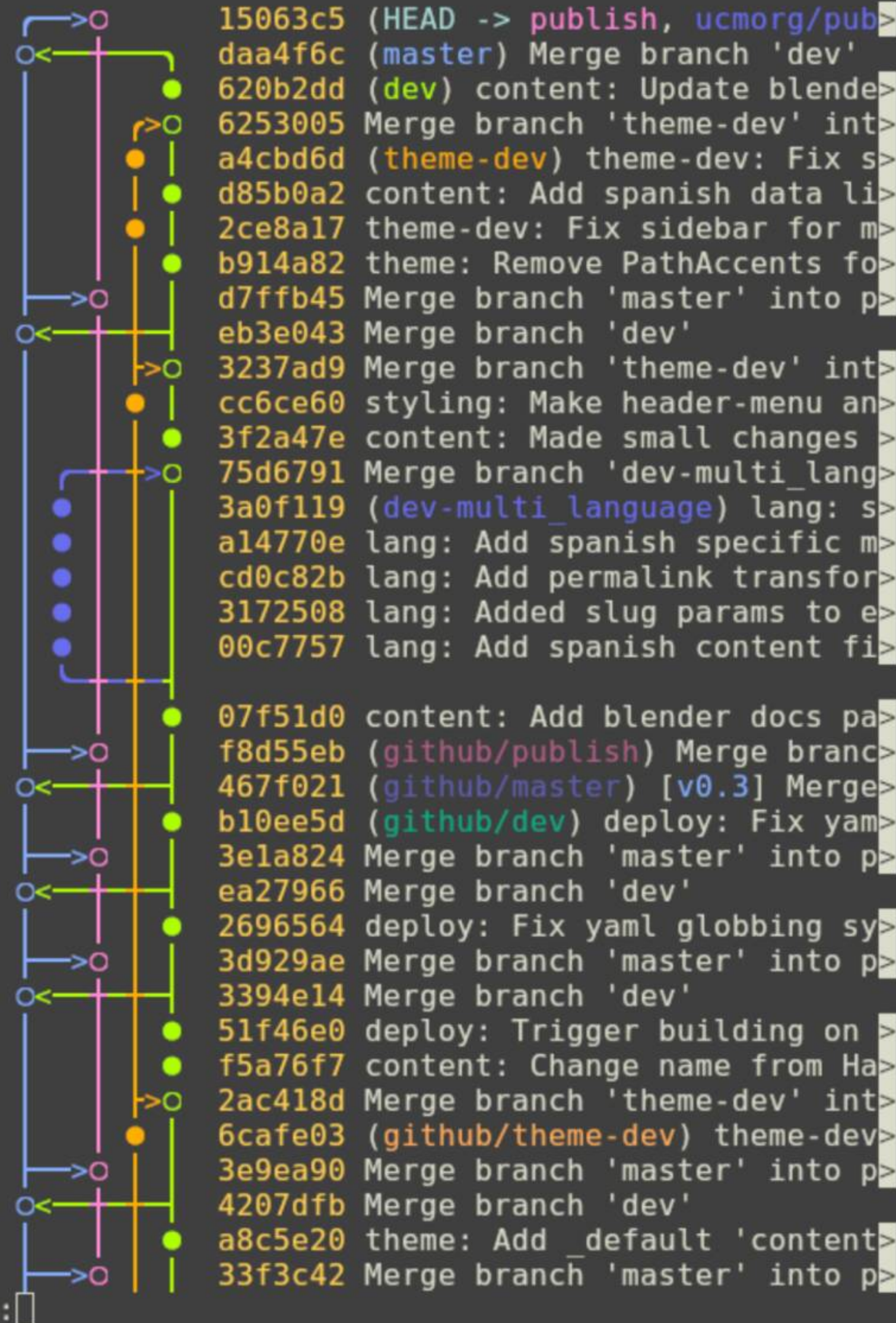


# Sys-Admin

\_I

- Bash
- Open-SHH
- Systemd
- Ansible
- LVM
- git
- vim
- ...

]\_



0[||| 3.1% 800MHz 31°C] Tasks: 111, 588 thr, 102 kthr; 1 run

1[||| 1.9% 400MHz 30°C] Load average: 0.48 0.20 0.21

2[||| 1.9% 400MHz 31°C] Uptime: 12 days, 14:55:05

3[||| 6.3% 400MHz 30°C]

Mem[||||| 3.26G/7.19G]

Swp[||||| 279M/512M]

| Main   |       | I/O  |    |       |       |      |   |      |      |         |           |  |  |
|--------|-------|------|----|-------|-------|------|---|------|------|---------|-----------|--|--|
| PID    | USER  | ΔPRI | NI | VIRT  | RES   | SHR  | S | CPU% | MEM% | TIME+   | Command   |  |  |
| 319    | root  | 20   | 0  | 31440 | 5324  | 2844 | S | 0.0  | 0.1  | 0:05.02 | /usr/lib/ |  |  |
| 1592   | root  | 20   | 0  | 15420 | 3916  | 3392 | S | 0.0  | 0.1  | 0:05.76 | /usr/lib/ |  |  |
| 3096   | root  | 20   | 0  | 8504  | 3576  | 3216 | S | 0.0  | 0.0  | 0:00.20 | sshd: /us |  |  |
| 3288   | root  | 20   | 0  | 313M  | 8880  | 4992 | S | 0.0  | 0.1  | 0:18.82 | /usr/lib/ |  |  |
| 3298   | root  | 20   | 0  | 313M  | 8880  | 4992 | S | 0.0  | 0.1  | 0:00.00 | /usr/lib/ |  |  |
| 3299   | root  | 20   | 0  | 313M  | 8880  | 4992 | S | 0.0  | 0.1  | 0:00.00 | /usr/lib/ |  |  |
| 3300   | root  | 20   | 0  | 313M  | 8880  | 4992 | S | 0.0  | 0.1  | 0:02.85 | /usr/lib/ |  |  |
| 3458   | root  | 20   | 0  | 2560  | 688   | 680  | S | 0.0  | 0.0  | 0:00.00 | fusermoun |  |  |
| 4831   | root  | 20   | 0  | 9452  | 920   | 784  | S | 0.0  | 0.0  | 0:00.03 | login --  |  |  |
| 28618  | root  | 20   | 0  | 456M  | 4732  | 3152 | S | 0.0  | 0.1  | 0:00.13 | /usr/lib/ |  |  |
| 28619  | root  | 20   | 0  | 456M  | 4732  | 3152 | S | 0.0  | 0.1  | 0:02.15 | /usr/lib/ |  |  |
| 28620  | root  | 20   | 0  | 456M  | 4732  | 3152 | S | 0.0  | 0.1  | 0:00.02 | /usr/lib/ |  |  |
| 28622  | root  | 20   | 0  | 456M  | 4732  | 3152 | S | 0.0  | 0.1  | 0:00.08 | /usr/lib/ |  |  |
| 28625  | root  | 20   | 0  | 456M  | 4732  | 3152 | S | 0.0  | 0.1  | 0:00.00 | /usr/lib/ |  |  |
| 28626  | root  | 20   | 0  | 456M  | 4732  | 3152 | S | 0.0  | 0.1  | 0:00.00 | /usr/lib/ |  |  |
| 34795  | root  | 20   | 0  | 176M  | 10088 | 3228 | S | 0.0  | 0.1  | 0:00.28 | /usr/bin/ |  |  |
| 149593 | root  | 20   | 0  | 154M  | 3912  | 3708 | S | 0.0  | 0.1  | 0:01.27 | /usr/bin/ |  |  |
| 228847 | root  | 20   | 0  | 17144 | 7632  | 5496 | S | 0.0  | 0.1  | 0:00.15 | wpa_suppl |  |  |
| 234112 | root  | 20   | 0  | 15852 | 10224 | 8688 | S | 0.0  | 0.1  | 0:00.06 | sshd-sess |  |  |
| 234162 | root  | 20   | 0  | 15852 | 10492 | 8700 | S | 0.0  | 0.1  | 0:00.03 | sshd-sess |  |  |
| 234374 | root  | 20   | 0  | 15852 | 10580 | 8916 | S | 0.0  | 0.1  | 0:00.03 | sshd-sess |  |  |
| 234560 | root  | 20   | 0  | 15852 | 10516 | 8724 | S | 0.0  | 0.1  | 0:00.04 | sshd-sess |  |  |
| 234763 | root  | 20   | 0  | 15156 | 6476  | 5580 | S | 0.0  | 0.1  | 0:00.01 | systemd-u |  |  |
| 234764 | root  | 20   | 0  | 15156 | 6500  | 5732 | S | 0.0  | 0.1  | 0:00.01 | systemd-u |  |  |
| 234765 | root  | 20   | 0  | 15156 | 6780  | 5884 | S | 0.0  | 0.1  | 0:00.01 | systemd-u |  |  |
| 3430   | rtkit | 21   | 1  | 21668 | 1444  | 1312 | S | 0.0  | 0.0  | 0:00.40 | /usr/lib/ |  |  |
| 3436   | rtkit | 20   | 0  | 21668 | 1444  | 1312 | S | 0.0  | 0.0  | 0:00.87 | /usr/lib/ |  |  |

wlp4s0 bmon 4.0

| Interfaces           | RX bps  | pps | % | TX bps  | pps | % |
|----------------------|---------|-----|---|---------|-----|---|
| lo                   | 0       | 0   |   | 0       | 0   |   |
| qdisc none (noqueue) | 0       | 0   |   | 0       | 0   |   |
| wlp4s0               | 6.50KiB | 11  |   | 2.69KiB | 6   |   |
| qdisc none (noqueue) | 0       | 0   |   | 0       | 0   |   |

MiB (RX Bytes/second)

KiB (TX Bytes/second)

28.63 112.23

23.85 93.53

19.08 74.82

14.31 56.12

9.54 37.41

4.77 18.71

1 5 10 15 20 25 30 35 40 45 50 55 60

1 5 10 15 20 25 30 35 40 45 50 55 60

Press d to enable detailed statistics

Press i to enable additional information

Fri Jan 31 12:03:28 2025

Press ? for help

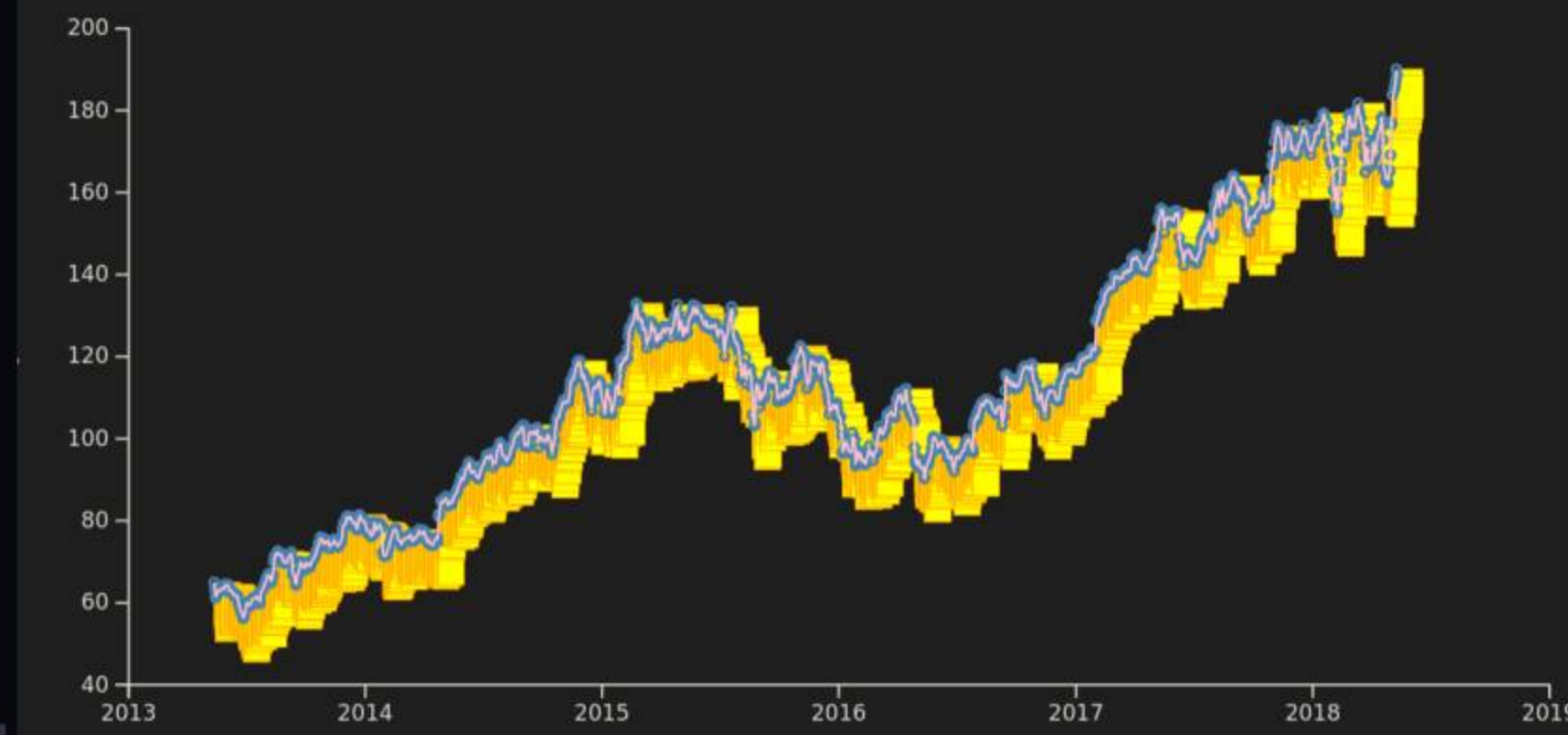
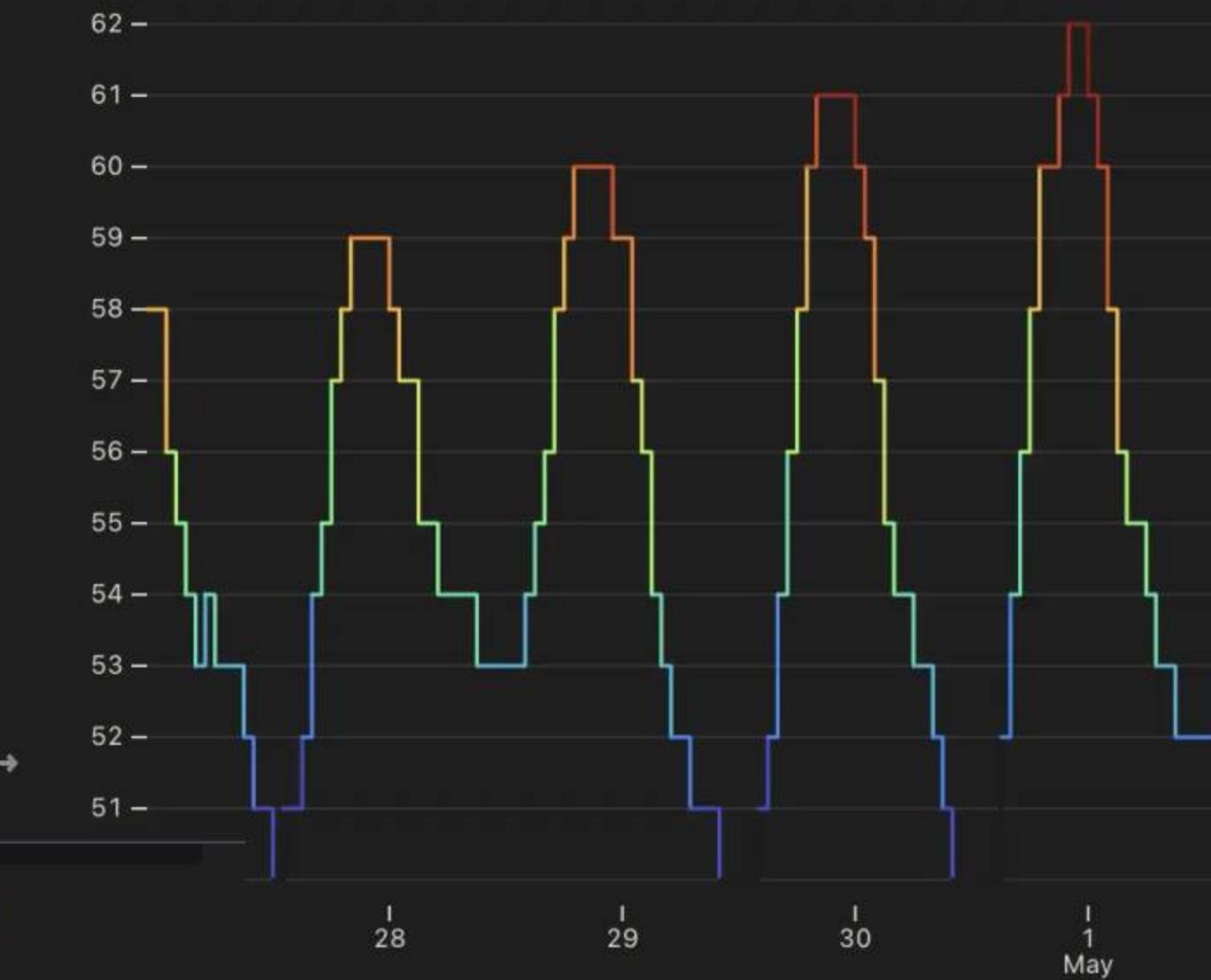


# Visualización de Datos

## Weather report

### Hourly temperature forecast

↑ Degrees (F)



► Array(9) [Object, Object, Object, Object, Object, Object, Object, Object, Object]

Search Query:

Number of Results:

## Wiki Data API



### The WikiData API

Let's have a look at what kind of information the WikiData page has on a particular artist: <https://www.wikidata.org/wiki/Q5582>

We can query the WikiData API for the information it has on Vincent Van Gogh. Van Gogh has an entity ID of Q5582.

```
curl 'https://www.wikidata.org/w/api.php?action=wbgetentities&ids=Q5582&p'
```

This gives us a bunch of different properties but too much to be useful to the naked eye without filtering. (Especially cause each claim/property is a wikimedia entity in itself.)

The 'work location' property has an property ID of P937.

So we can filter this further with the jq command:

```
curl 'https://www.wikidata.org/w/api.php?action=wbgetentities&ids=Q5582&p'
```



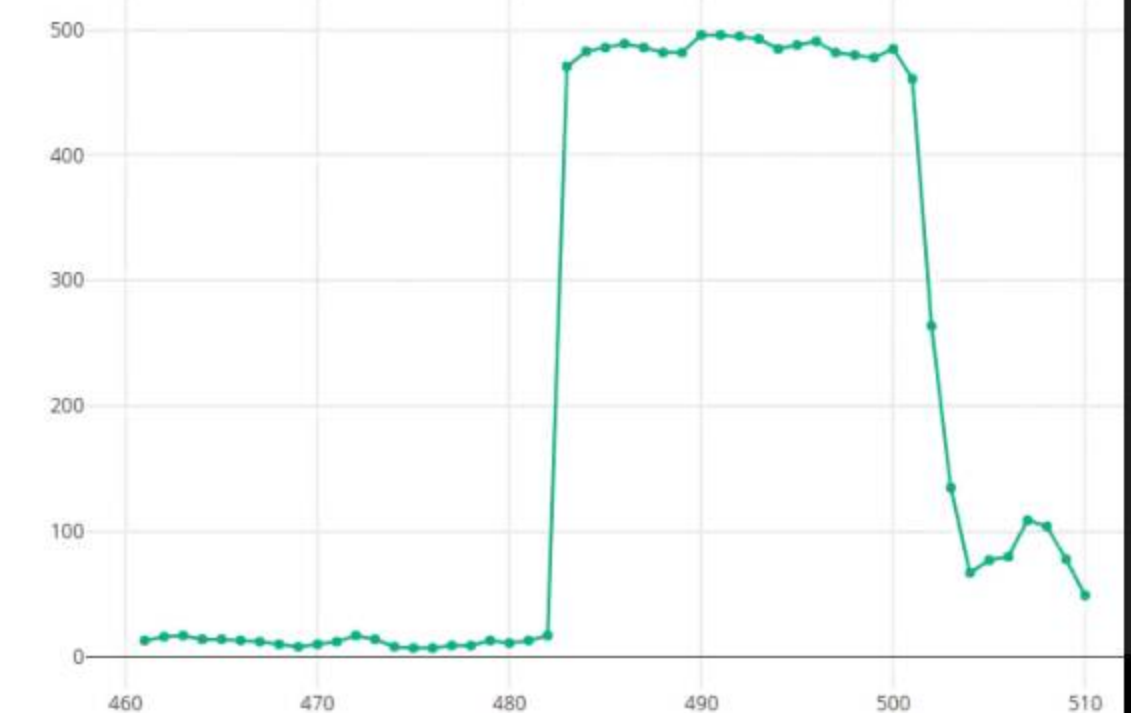
ID: 436964  
1866 - 1866  
Title: Young Lady in 1866  
Artist: Edouard Manet  
Object Name: Painting  
Department: European Paintings

## MicroPython WebSocket

### Real time values

49  
78  
104  
109  
80  
77  
67  
135  
264  
461  
485  
478  
480  
482  
491  
488  
485  
493  
495  
496

### Capacitive Touch-Probe



WebSocket message received:

```
► MessageEvent {isTrusted: true, data: '104', origin: 'ws://192.168.1.111', lastEventId: '', source: null, ...}
```

WebSocket message received:

```
► MessageEvent {isTrusted: true, data: '78', origin: 'ws://192.168.1.111', lastEventId: '', source: null, ...}
```

WebSocket message received:

```
► MessageEvent {isTrusted: true, data: '49', origin: 'ws://192.168.1.111', lastEventId: '', source: null, ...}
```